# Mathematics Exercises

2026

# ESQC mathematics exercises

**2026 edition**

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# Preface

This is a collection of mathematics exercises for the European Summerschool in Quantum Chemistry (ESQC).

# Part I

# Exercise recommendations

# Recommended exercises for the first day

This chapter contains recommended exercises for the mathematics tutorial on the first day of the school. These recommended exercises are considered especially useful.

We provide also a list of recommended exercises for the 'beginners'. If you feel that you need training in the *absolute basics* of the mathematics used in quantum chemistry, we recommend that you start here.

In the next chapter, we provide lists of recommended exercises for each quantum chemistry topic taught at the school.

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# Recommended exercises by topic

This chapter contains lists of mathematics exercises grouped by each quantum chemistry topic taught at the school. These exercises can be useful as a warm-up to the proper exercises given each day.

## Coupled-cluster theory

## Second quantization

## Self-consistent field theory

## Density functional theory

## Response theory

## Relativistic quantum chemistry

## Molecular properties

## Multiconfigurational methods

# Part II

# Mathematical foundations

# 1 Sets and functions

Sets and set notation appear many places in quantum chemistry. The goal of this chapter is to train you in the basic usage of sets and set notation.

## 1.1 Exercises: abstract sets and functions

**Exercise 1.1.** Write the days of the week as a set

**Exercise 1.2.** Write the five first natural numbers using set notation as indicated in the lecture notes

**Exercise 1.3.** An *ordered pair* $(x, y)$ is a two-element "set" where the order matters. Two ordered pairs $(x, y)$ and $(u, v)$ are equal if and only if $x = u$ and $y = v$. From this it follows that $(x, y) = (y, x)$ if and only if $x = y$. A definition of an ordered pair as a *set* is

$$(x, y) = \{\{x\}, \{x, y\}\}.$$

Show that $(x, y) = (y, x)$ if and only if $x = y$ using the set theoretic definition.

**Exercise 1.4.** A nonnegative rational number $p/q$, $p, q \in \mathbb{N}$, $q > 0$, can be written as an ordered pair $(p, q)$. Write down the rational numbers $1/4$, $2/3$, $0$, $1/3$ as ordered pairs using set theory, both for the pair and for each integer. (Strictly speaking we the number itself is an *equivalence class* of such pairs, i.e., $pn/qn = p/q$, so we must identify $(p, q)$ and $(np, nq)$. Ignore equivalence classes in this exercise.)

**Exercise 1.5.** The *cartesian product* of two sets $A$ and $B$ is

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

Write down the cartesian product of $\{\heartsuit, \diamondsuit, \spadesuit, \clubsuit\}$ and $\{1, 2, 3\}$, using the set theoretic definition for the ordered pair. You can skip spelling out the set theoretic definition of the natural numbers.

**Exercise 1.6.** Similar to the previous exercise, write down $\{1, 2, 3\} \times \{2, 3, 4\}$.

**Exercise 1.7.** A *function* $f : A \to B$ from one set $A$ to another set $B$ is a rule that assigns to every $a \in A$ precisely one $b \in B$. In terms of set theory, a function $f : A \to B$ is a subset of $A \times B$, such that

- For all $a \in A$ there exists $b \in B$ such that $(a, b) \in f$.
- For all $a \in A$ and $b, b' \in B$, if $(a, b) \in f$ and $(a, b') \in f$, then $b = b'$.

Write down the function $f : \{1, 2, 3\} \to \{1, 2, 3\}$, $f(1) = 2$, $f(2) = 3$, $f(3) = 1$, using the set theoretic definition of ordered pairs and the cartesian product.

De Morgan's laws are useful when discussing subsets $A, B$ of a larger set $X$. Recall that the complement of $A$ relative to $X$ is $A^{\complement} = X \setminus A$. De Morgan's laws state that:

- $(A \cup B)^{\complement} = A^{\complement} \cap B^{\complement}$.
- $(A \cap B)^{\complement} = A^{\complement} \cup B^{\complement}$.

**Exercise 1.8.** Draw a picture, representing $X$ as the whole sheet, $A$ and $B$ as overlapping shapes, e.g., circles. Label $A$, $B$, $A \cap B$, and $A \cup B$, making another drawing if necessary. Convince yourself that De Morgan's laws are correct.

**Exercise 1.9.** Prove De Morgan's laws mathematically. Note that the complement operation acts like negation of truth, i.e., $a \in A^{\complement}$ if and only if $a \in X$ and $a \notin A$.

## 1.2 Exercises: Sets in quantum chemistry

**Exercise 1.10.** Figure Figure 1.1 shows the *Morse potential* - a common model potential for vibrational motion of a diatomic molecule:

$$V(r) = D_e \left(1 - e^{-a(r-r_e)}\right)^2 - D_e,$$

where $D_e > 0$ is the dissociation energy, $r_e > 0$ is the equilibrium bond length, and $a > 0$ is a parameter that determines the width of the potential well. The total energy of the molecule is denoted by $E \in \mathbb{R}$. The classically allowed region for the total energy $E$ is the set $S_E$ of all bond lengths $r$ such that $V(r) \leq E$:

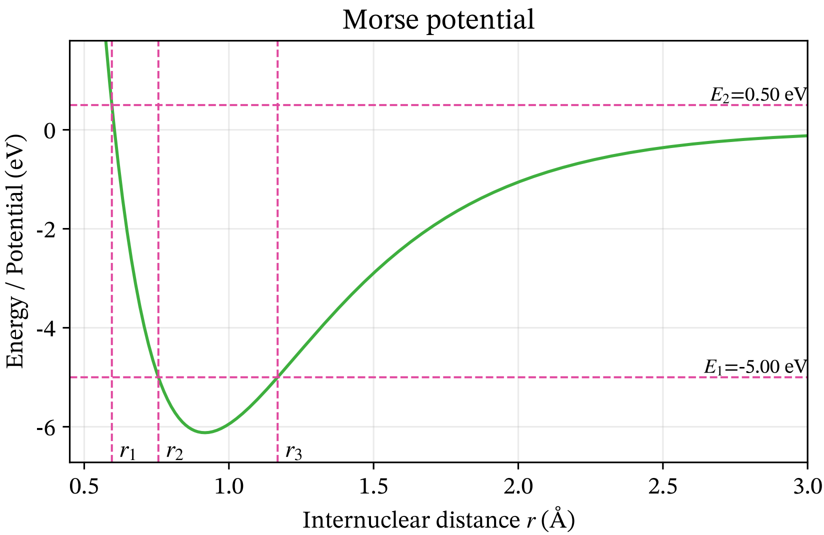



Figure 1.1: Morse potential

$$S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$$

a. Explain the notation $S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$ in words. In particular, what does the vertical bar $|$ mean?

b. Can you find a condition on $E$ such that $S_E$ is the empty set?

c. Can you find a condition on $E$ such that $S_E$ is the whole real line?

d. Based on the figure, which of the following statements are true?

- $S_{E_1} = (r_2, r_3)$
- $S_{E_1} = [r_2, r_3]$
- $S_{E_1} = \{r_2, r_3\}$

e. Based on the figure, which of the following statements are true?

- $S_{E_2} = (r_1, +\infty)$
- $S_{E_2} = [r_1, +\infty)$
- $S_{E_2} = \{r_1, r_2, r_3\}$

**Exercise 1.11.** In this exercise, a *molecule* is informally defined only by its chemical composition: the number of atoms of each element occurring in its molecular formula. Thus, $H_2$ and CO are two distinct molecules, while HCOOH and $CO_2H_2$ represent the same molecule, since both contain one C atom, two H atoms, and two O atoms.

We ignore molecular geometry, bonding, isotopes, charge, and all questions of chemical stability. In particular, any formal combination of atoms is deemed a molecule, whether or not such a species could exist physically.

We restrict our attention to the four elements C, H, O, and N. Let $M$ denote the set of all molecules that can be constructed from these elements. A molecule must contain at least one atom.

One useful way of describing a molecule is by the ordered quadruple

$$(n_{\mathrm{C}}, n_{\mathrm{H}}, n_{\mathrm{O}}, n_{\mathrm{N}}), \tag{1.1}$$

where each $n_X$ is the number of atoms of element $X$ in the molecule. For example,

$$\mathrm{CH_4} \longleftrightarrow (1, 4, 0, 0),$$

and

$$\mathrm{N_2O} \longleftrightarrow (0, 0, 1, 2).$$

a. Explain why Equation 1.1 uniquely determines an element of $M$. Give 4-5 additional examples of molecules in terms of this representation.

b. Express $M$ in terms of $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. What is the cardinality of $M$? Is it finite, countably infinite, or uncountably infinite? Justify your answer.

c. Let

$$M_2 = \{m \in M \mid m \text{ contains exactly two atoms}\}.$$

Find $|M_2|$ and list all elements of $M_2$.

d. Let $S \subset M$ be the set

$$S = \{m \in M \mid m \text{ contains at least one H atom}\}.$$

Describe $S$ in terms of the quadruple representation above. Is $S$ finite, countably infinite, or uncountably infinite? Justify your answer.

d. Describe the complement $M \setminus S$. What is its cardinality?

e. Define

$$C = \{m \in M \mid m \text{ contains at least one C atom}\}.$$

Describe the following sets both in words and in terms of the quadruple representation:

$$C \cap S, \qquad C \cup S, \qquad C \setminus S.$$

Determine the cardinality of each set.

f. For $n \geq 1$, define

$$M_n = \{m \in M \mid m \text{ contains exactly } n \text{ atoms}\}.$$

Thus, a molecule in $M_n$ satisfies

$$n_{\mathrm{C}} + n_{\mathrm{H}} + n_{\mathrm{O}} + n_{\mathrm{N}} = n.$$

Show that

$$|M_n| = \binom{n+3}{3}.$$

Check that your formula agrees with your answer to part **b**.

g. Explain why

$$M = \bigcup_{n=1}^{\infty} M_n.$$

Are the sets $M_n$ pairwise disjoint? Use this decomposition to give another argument that $M$ is countably infinite.

h. Show that $M$ and $S$ have the same cardinality. Explain why this is possible even though $S$ is a proper subset of $M$.

## 1.3 Solutions

### 1.3.1 Tutor note: Sets etc.

The first two exercises should be accessible to everyone. The von Neumann representation of natural numbers grows quickly, so five numbers suffice.

The visual exercise on De Morgan's laws supports the later discussion of open sets and is recommended.

#### Solution to Exercise 1.1

$$U = \{\text{Mandag, Tirsdag, Onsdag, Torsdag, Fredag, Lørdag, Søndag}\}$$

#### Solution to Exercise 1.2

Okay, so each number doubles in size. After ten numbers, we have on the order of 1000 characters per number. So the five first is enough.

With $0 = \varnothing$ and $n + 1 = n \cup \{n\}$, the first five natural numbers are

$$
\begin{aligned}
0 &= \varnothing, \\
1 &= \{\varnothing\}, \\
2 &= \{\varnothing, \{\varnothing\}\}, \\
3 &= \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}, \\
4 &= \{0, 1, 2, 3\}.
\end{aligned}
$$

### Solution to Exercise 1.3

*The source does not supply a solution.*

### Solution to Exercise 1.4

*The source does not supply a solution.*

### Solution to Exercise 1.5

*The source does not supply a solution.*

### Solution to Exercise 1.6

*The source does not supply a solution.*

### Solution to Exercise 1.7

*The source does not supply a solution.*

### Solution to Exercise 1.8

*The source does not supply a solution.*

## Solution to Exercise 1.9

Here is the proof of the first one: $x \in (A \cup B)^{\complement}$ iff $x \notin A \cup B$ iff $x \notin A$ and $x \notin B$ iff $x \in A^{\complement}$ and $x \in B^{\complement}$ iff $x \in A^{\complement} \cap B^{\complement}$.

## Solution to Exercise 1.10

a. The notation

$$S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$$

means that $S_E$ is the set of all real numbers $r$ for which the potential energy $V(r)$ is less than or equal to the total energy $E$.

The vertical bar | means **such that**. Thus, the expression can be read as

> the set of all $r$ in the real numbers such that $V(r) \leq E$.

b. The Morse potential has its minimum at the equilibrium bond length $r = r_e$. At this point,

$$V(r_e) = D_e(1 - e^0)^2 - D_e = -D_e.$$

Therefore,

$$V(r) \geq -D_e$$

for all $r$. If

$$E < -D_e,$$

there are no values of $r$ for which $V(r) \leq E$. Hence,

$$\boxed{S_E = \varnothing \quad \text{if} \quad E < -D_e.}$$

Notice that when $E = -D_e$, the set is not empty; it contains exactly the equilibrium bond length:

$$S_{-D_e} = \{r_e\}.$$

c. There is no finite value of $E$ for which $S_E$ is the whole real line.

To see this, consider the limit as $r \to -\infty$. Since $a > 0$,

$$e^{-a(r-r_e)} \to +\infty,$$

and therefore

$$V(r) \to +\infty.$$

Thus, for any finite $E \in \mathbb{R}$, there are sufficiently negative values of $r$ for which

$$V(r) > E.$$

Consequently,

$$\boxed{S_E \neq \mathbb{R} \quad \text{for every finite } E \in \mathbb{R}.}$$

d. At energy $E_1$, the horizontal energy line intersects the potential at the two turning points $r_2$ and $r_3$. Between these points,

$$V(r) \leq E_1.$$

The endpoints themselves are included because

$$V(r_2) = V(r_3) = E_1$$

and the definition uses $\leq$, rather than $<$.

Therefore,

$$\boxed{S_{E_1} = [r_2, r_3].}$$

Hence the statements have truth values

- $S_{E_1} = (r_2, r_3)$: **false**

- $S_{E_1} = [r_2, r_3]$: **true**
- $S_{E_1} = \{r_2, r_3\}$: **false**

e. At energy $E_2$, the potential first becomes less than or equal to $E_2$ at $r = r_1$. For all larger $r$, the potential remains below or equal to $E_2$.

Again, the point $r_1$ itself is included because

$$V(r_1) = E_2.$$

Therefore,

$$\boxed{S_{E_2} = [r_1, +\infty).}$$

Hence the statements have truth values

- $S_{E_2} = (r_1, +\infty)$: **false**
- $S_{E_2} = [r_1, +\infty)$: **true**
- $S_{E_2} = \{r_1, r_2, r_3\}$: **false**

### Solution to Exercise 1.11

a. A molecule is uniquely determined by an ordered quadruple

$$(n_{\mathrm{C}}, n_{\mathrm{H}}, n_{\mathrm{O}}, n_{\mathrm{N}}),$$

where each entry is a non-negative integer and not all four entries are zero. Hence

$$M = \mathbb{N}_0^4 \setminus \{(0, 0, 0, 0)\}.$$

Since $\mathbb{N}_0$ is countably infinite, any finite Cartesian product of copies of $\mathbb{N}_0$ is also countably infinite. Removing one element does not change the cardinality. Therefore

$$|M| = \aleph_0,$$

so $M$ is **countably infinite**.

b. A diatomic molecule corresponds to a non-negative integer solution of

$$n_{\mathrm{C}} + n_{\mathrm{H}} + n_{\mathrm{O}} + n_{\mathrm{N}} = 2.$$

The possibilities are

$$\mathrm{C}_2, \quad \mathrm{H}_2, \quad \mathrm{O}_2, \quad \mathrm{N}_2, \quad \mathrm{CH}, \quad \mathrm{CO}, \quad \mathrm{CN}, \quad \mathrm{HO}, \quad \mathrm{HN}, \quad \mathrm{ON}.$$

Hence

$$|M_2| = 10.$$

c. In the quadruple representation,

$$S = \left\{(n_{\mathrm{C}}, n_{\mathrm{H}}, n_{\mathrm{O}}, n_{\mathrm{N}}) \in \mathbb{N}_0^4 \mid n_{\mathrm{H}} \geq 1\right\}.$$

The set $S$ is infinite, since it contains

$$\mathrm{H}, \mathrm{H}_2, \mathrm{H}_3, \ldots$$

It is also a subset of the countable set $M$. Therefore

$$|S| = \aleph_0,$$

so $S$ is countably infinite.

d. The complement consists of all molecules containing no hydrogen:

$$M \setminus S = \{(n_{\mathrm{C}}, 0, n_{\mathrm{O}}, n_{\mathrm{N}}) \mid n_{\mathrm{C}}, n_{\mathrm{O}}, n_{\mathrm{N}} \in \mathbb{N}_0,\ n_{\mathrm{C}} + n_{\mathrm{O}} + n_{\mathrm{N}} > 0\}.$$

This set is again countably infinite:

$$|M \setminus S| = \aleph_0.$$

e. We have

$$C = \{(n_\mathrm{C}, n_\mathrm{H}, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1\}.$$

The intersection

$$C \cap S = \{(n_\mathrm{C}, n_\mathrm{H}, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1,\ n_\mathrm{H} \geq 1\}$$

consists of molecules containing both carbon and hydrogen.

The union

$$C \cup S = \{(n_\mathrm{C}, n_\mathrm{H}, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1 \text{ or } n_\mathrm{H} \geq 1\}$$

consists of molecules containing carbon or hydrogen, or both.

Finally,

$$C \setminus S = \{(n_\mathrm{C}, 0, n_\mathrm{O}, n_\mathrm{N}) \in M \mid n_\mathrm{C} \geq 1\}$$

consists of molecules containing carbon but no hydrogen.

All three sets are infinite subsets of the countable set $M$, and each is easily seen to contain a countably infinite sequence of distinct molecules. Hence

$$|C \cap S| = |C \cup S| = |C \setminus S| = \aleph_0.$$

f. The number of elements of $M_n$ is the number of non-negative integer solutions of

$$n_\mathrm{C} + n_\mathrm{H} + n_\mathrm{O} + n_\mathrm{N} = n.$$

By the stars-and-bars formula, the number of such solutions is

$$|M_n| = \binom{n+4-1}{4-1} = \binom{n+3}{3}.$$

For $n = 2$,

$$|M_2| = \binom{5}{3} = 10,$$

in agreement with part **b**.

g. Every molecule contains some finite positive number $n$ of atoms, so it belongs to exactly one set $M_n$. Therefore

$$M = \bigcup_{n=1}^{\infty} M_n.$$

The sets $M_n$ are pairwise disjoint, since a molecule cannot contain exactly $n$ atoms and exactly $m$ atoms when $n \neq m$.

Each $M_n$ is finite, since

$$|M_n| = \binom{n+3}{3}.$$

Thus $M$ is a countable union of finite sets. Therefore $M$ is countable. Since there are infinitely many molecules, $M$ is countably infinite.

h. From parts **a** and **c**,

$$|M| = |S| = \aleph_0.$$

Thus $M$ and $S$ have the same cardinality, even though

$$S \subsetneq M.$$

This is possible because $M$ is infinite. Unlike finite sets, an infinite set can have the same cardinality as one of its proper subsets.

For example, enumerate the molecules as

$$M = \{m_1, m_2, m_3, \dots\}$$

and the hydrogen-containing molecules as

$$S = \{s_1, s_2, s_3, \ldots\}.$$

Then the map

$$f : M \to S, \qquad f(m_i) = s_i,$$

is a bijection. Hence $M$ and $S$ have the same cardinality.

# 2 Cardinality and countability

Cardinality compares sets through bijections. The exercises cover countable sets, Cantor's diagonal argument, and the cardinality of the continuum.

Recall that the *cardinality* $|S|$ of a set $S$ is the number of elements in $S$. For finite sets, this is intuitive, but what about infinite sets?

One says that $A$ and $B$ have the same cardinality if there exists a bijection $f : A \to B$, i.e., $f$ is one-to-one and onto. (Intuitively, you can draw lines between individual elements of $A$ and $B$, only one line to/from each element. Precisely: Onto means, to every $b \in B$ there exists a $a \in A$ such that $f(a) = b$. One-to-one: $f(a) = f(a')$ implies $a = a'$.) One can think of the bijection as labelling each $b \in B$ by exactly one element $a \in A$, and that all labels from $A$ are used up.

Let $\mathbb{N}_n = \{1, 2, \cdots, n\}$. This set has cardinality $n$.

The set $\mathbb{N}$ has an infinite number of elements. By definition, $|\mathbb{N}| = \aleph_0$ ("aleph-nought").

## 2.1 Exercises

**Exercise 2.1.** Show that $|\mathbb{Z}| = \aleph_0$ by explicitly constructing a bijection $f : \mathbb{N} \to \mathbb{Z}$.

**Exercise 2.2.** Show that $|\mathbb{N} \times \mathbb{N}| = \aleph_0$. Hint: Draw a picture of $\mathbb{N} \times \mathbb{N}$, and try to draw a line through all the points in this set. How does this show the existence of a bijection between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$?

**Exercise 2.3.** Show that $|\mathbb{Q}| = \aleph_0$.

**Exercise 2.4.** Show that the interval $I = (-1, 1)$ has the same cardinality as $\mathbb{R}$. You must construct a function $f : (-1, 1) \to \mathbb{R}$ that is a bijection.

**Exercise 2.5.** In this exercise, we show that $|\mathbb{R}| > \aleph_0$. The cardinality would be $\aleph_0$ if we could write a list of the real numbers. It suffices to find a list of the numbers in $I = [0, 1[$, i.e., infinite decimal expansions $0.d_1 d_2 d_3 \cdots$, with $d_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

We will look for a contradiction. Consider a list $x_1, x_2, x_3, \ldots$ of all real numbers in $[0, 1[$. This is essentially an infinite matrix of digits. Can you find a real number $x \in [0, 1]$ which is *not* in this list, i.e., $x \neq x_j$ for all $j$? Hint: consider the diagonal of the matrix.

The cardinality of $\mathbb{R}$, "the continuum", is written $2^{\aleph_0}$.

**Exercise 2.6.** Show that $|[0, 1[\times[0, 1[| = |[0, 1[|$, by constructing a one-to-one map. Hint: Use decimal expansions. Conclude that also $|\mathbb{R}^2| = |\mathbb{C}| = |\mathbb{R}|$.

## 2.2 Solutions

### Solution to Exercise 2.1

We construct a map $f : \mathbb{N} \to \mathbb{Z}$. We let the even numbers in $\mathbb{N}$ map to positive numbers in $\mathbb{Z}$, and odd numbers in $\mathbb{N}$ to negative numbers. For every $n \in \mathbb{N}$,

$$f(2n) = n, \quad f(2n + 1) = -n - 1.$$

We get $f(0) = 0$, $f(1) = -1$, $f(2) = 1$, $f(3) = -2$, etc.

### Solution to Exercise 2.2

Draw a picture of $\mathbb{N}\times\mathbb{N}$ - a uniform grid. The line can be drawn in many ways, but the easiest is perhaps to connect $(0, 0)-(0, 1)-(1, 0)-(2, 0)-(1, 1)-(0, 2)-\ldots$

### Solution to Exercise 2.3

Use that $x \in \mathbb{Q}$ is represented as a fraction, i.e., a point in $\mathbb{N} \times \mathbb{N}$. Actually, $x$ is an equivalence class, so some points in $\mathbb{N} \times \mathbb{N}$ are eliminated.

## Solution to Exercise 2.4

One possible bijection is

$$f(x) = \frac{x}{1 - |x|}, \qquad -1 < x < 1.$$

Its inverse is $g(y) = y/(1 + |y|)$ for $y \in \mathbb{R}$.

## Solution to Exercise 2.5

The proof is called Cantor's diagonalization argument, and is used in several other important results. (For example, to prove that there is no Turing machine that can determine whether an arbitrary Turing machine stops.)

Let the $k$th digit be different from the $k$th digit of number $k$ in the list. Then the real number thus constructed cannot be equal to any of the numbers in the list.

## Solution to Exercise 2.6

Let every even-numbered decimal from $x$ be the decimal of $y_1$ and odd decimal the decimal of $y_2$, respectively. This maps decimal expansions $x$ onto pairs of decimal expansions $(y_1, y_2)$. Clearly, the inverse mapping can be created, too.

# 3 Combinatorics

Combinatorics is the part of mathematics concerned with *counting* in a broad sense, for example counting the number of ways to arrange a set of objects or the number of combinations that can be made from a set of objects. It is a fundamental part of discrete mathematics and has applications in many areas of mathematics, computer science, chemistry, and physics.

## 3.1 Exercises: Factorials

The *factorial* of a non-negative integer $n$ is defined as the product of all positive integers less than or equal to $n$:

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1, \quad n \geq 1.$$

It is also defined that $0! = 1$. The factorial appears in many mathematical contexts, including series expansions, and of course combinatorics. Its basic use is the following:

> **Factorial and counting**
>
> Given a set of $n$ distinct objects, the number of ways to arrange them in a sequence (i.e., the number of permutations) is $n!$.

In one exercise, you will need the following approximation for large $n$, known as Stirling's approximation:

> **Stirling's approximation**
>
> For large $n$, the factorial $n!$ can be approximated as

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

In calculations, especially numerical calculations, it is often convenient to use the logarithm of the factorial. Using Stirling's approximation, we have

$$\log(n!) \approx \frac{1}{2}\log(2\pi n) + n\log(n) - n.$$

**Exercise 3.1.** Compute the factorials 0!, 1!, 2!, 3!, 4!, and 5!.

**Exercise 3.2.** 42 people stand in a lunch line. How many different lunch line arrangements with 42 people are possible?

**Exercise 3.3.** Compute the number of arrangements for the following scenarios:

a. A chemist has 10 different chemical samples. How many ways can the chemist arrange the samples on a shelf?
b. A chemist has 10 different chemical samples. 4 of the samples are liquids, and the other 6 are solids. How many ways can the chemist arrange the samples on a shelf if all the liquids must be together?

**Exercise 3.4** (Stirling's approximation)**.** We study Stirling's approximation for the factorial.

a. Use Stirling's approximation to estimate 5!. Compare your result with the exact result

$$5! = 120.$$

b. Use Stirling's approximation to estimate 10!. Compare again with the exact number

$$10! = 3\,628\,800.$$

c. The relative error of an approximation $A$ to an exact value $x$ is

$$\frac{|A-x|}{x}.$$

Calculate the relative error of Stirling's approximation for 5! and 10!. For which value of $n$ is the approximation better?

d. Use Stirling's approximation to estimate 50!. The exact value is approximately

$$50! \approx 3.041 \times 10^{64}.$$

e. Based on your results above, what happens to the accuracy of Stirling's approximation as $n$ increases?

## 3.2 Exercises: Binomial coefficients and Pascal's triangle

The *binomial coefficient* $\binom{n}{k}$ is defined as the number of ways to choose $k$ elements from a set of $n$ elements, without regard to the order of selection. It is given by the formula:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

Pascal's triangle is a triangular array of numbers where each number is the sum of the two numbers directly above it:

$$1 \tag{3.1}$$
$$1 \quad 1 \tag{3.2}$$
$$1 \quad 2 \quad 1 \tag{3.3}$$
$$1 \quad 3 \quad 3 \quad 1 \tag{3.4}$$
$$1 \quad 4 \quad 6 \quad 4 \quad 1 \tag{3.5}$$
$$\vdots \tag{3.6}$$

Fact: The $n$-th row of Pascal's triangle contains the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

**Exercise 3.5.**

a. Compute the binomial coefficients $\binom{5}{2}$, $\binom{6}{3}$, and $\binom{7}{4}$.

b. In Norwegian Lotto, 7 numbers are drawn from a set of 34 numbers. How many different combinations of 7 numbers can be drawn?

**Exercise 3.6.** Compute the first 6 rows of Pascal's triangle. Verify that the numbers in the $n$-th row correspond to the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

**Exercise 3.7** (Pascal's identity)**.** The binomial coefficients satisfy **Pascal's identity**,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

a. Explain why this relation is equivalent to the construction of Pascal's triangle, where each number is the sum of the two numbers directly above it.

b. Write each binomial coefficient on the right-hand side in terms of factorials.

c. Put the two terms over a common denominator and simplify the expression.

d. Show that the resulting expression is

$$\frac{n!}{k!(n-k)!},$$

and hence prove Pascal's identity.

e. Give a combinatorial explanation of Pascal's identity. *Hint:* Consider choosing $k$ objects from a set of $n$ objects. Pick one particular object and separate the choices into two cases: those that contain this object and those that do not.

**Exercise 3.8.** Write a computer program to generate the first $2^n$ rows of Pascal's triangle, where $n = 1, 2, \dots$. Create a $2^n \times 2^n$ lower triangular matrix $T$, where $T_{k,n} = 0$ if $\binom{n}{k}$ is even, and $T_{k,n} = 1$ if the coefficient is odd, for $k \leq n$ (indexing from zero). Visualize the matrix. What pattern do you observe? This pattern is known as the Sierpinski triangle.

If you use Python and sympy, you can generate binomial factors with the function `sympy.binomial(n, k)`. Check for parity using the modulo operator `%`. For example, `sympy.binomial(n, k) % 2` will return `0` for even numbers and `1` for odd numbers.

## 3.3 Solutions

### Solution for Exercise 3.1

Recall that $0! = 1$ and, for a positive integer $n$,

$$n! = n(n-1)\cdots 2 \cdot 1.$$

Therefore,

$$0! = 1, \qquad 1! = 1, \qquad 2! = 2, \qquad 3! = 6, \qquad 4! = 24, \qquad 5! = 120.$$

### Solution for Exercise 3.2

There are 42 choices for the first person in the line, 41 choices for the second, and so on. Hence the number of possible arrangements is

$$42! = 42 \cdot 41 \cdots 2 \cdot 1.$$

### Solution for Exercise 3.3

a. The 10 different chemical samples can be arranged in

$$10! = 3\,628\,800$$

different ways.

b. Since the 4 liquids must be together, we can first regard them as a single object. We then have this object together with the 6 solids, giving 7 objects to arrange. These can be arranged in 7! ways. For each such arrangement, the 4 liquids can be arranged among themselves in 4!. Therefore, the total number of arrangements is

$$7!4! = 120\,960.$$

## Solution for Exercise 3.4

We use Stirling's approximation,

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

a. For $n = 5$,

$$5! \approx \sqrt{10\pi}\left(\frac{5}{e}\right)^5 \approx 118.0.$$

The exact value is

$$5! = 120.$$

Thus, Stirling's approximation is already quite close for $n = 5$.

b. For $n = 10$,

$$10! \approx \sqrt{20\pi}\left(\frac{10}{e}\right)^{10} \approx 3.599 \times 10^6.$$

The exact value is

$$10! = 3.6288 \times 10^6.$$

Again, the approximation is very close to the exact value.

c. For 5!, the relative error is

$$\frac{|118.0 - 120|}{120} \approx 0.0165,$$

or about

$$1.65\%.$$

For 10!, the relative error is

$$\frac{|3.599 \times 10^6 - 3.6288 \times 10^6|}{3.6288 \times 10^6} \approx 0.0083,$$

or about

$$0.83\%.$$

Thus, the approximation is better for $n = 10$ than for $n = 5$.

d. For $n = 50$,

$$50! \approx \sqrt{100\pi}\left(\frac{50}{e}\right)^{50} \approx 3.036 \times 10^{64}.$$

For comparison, the exact value is approximately

$$50! \approx 3.041 \times 10^{64}.$$

e. The relative error becomes smaller as $n$ increases. Thus, Stirling's approximation becomes increasingly accurate for large $n$.

## Solution for Exercise 3.5

a. The binomial coefficient is given by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Thus,

$$\binom{5}{2} = \frac{5!}{2!3!} = \frac{5 \cdot 4}{2 \cdot 1} = 10,$$

$$\binom{6}{3} = \frac{6!}{3!3!} = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = 20,$$

and

$$\binom{7}{4} = \frac{7!}{4!3!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

b. Since the order in which the 7 numbers are drawn does not matter, the number of possible combinations is

$$\binom{34}{7} = \frac{34!}{7!27!} = \frac{34 \cdot 33 \cdot 32 \cdot 31 \cdot 30 \cdot 29 \cdot 28}{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 5\,379\,616.$$

Thus, there are **5 379 616** different combinations of 7 numbers.

## Solution for Exercise 3.7

a. The relation is equivalent to the construction of Pascal's triangle because each number in the triangle is the sum of the two numbers directly above it. The left term on the right-hand side corresponds to the number directly above and to the left, while the right term corresponds to the number directly above and to the right:

$$\binom{0}{0} \tag{3.7}$$
$$\binom{1}{0} \quad \binom{1}{1} \tag{3.8}$$
$$\binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \tag{3.9}$$
$$\binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3} \tag{3.10}$$
$$\vdots \tag{3.11}$$

b. Using

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

we have

$$\binom{n-1}{k-1} = \frac{(n-1)!}{(k-1)!(n-k)!}$$

and

$$\binom{n-1}{k} = \frac{(n-1)!}{k!(n-k-1)!}.$$

c. We write both terms with the common denominator

$$k!(n-k)!.$$

Thus,

$$\binom{n-1}{k-1} = \frac{k(n-1)!}{k!(n-k)!},$$

and

$$\binom{n-1}{k} = \frac{(n-k)(n-1)!}{k!(n-k)!}.$$

Therefore,

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{k(n-1)! + (n-k)(n-1)!}{k!(n-k)!}.$$

d. Factoring out $(n-1)!$ gives

$$\frac{[k+(n-k)](n-1)!}{k!(n-k)!} = \frac{n(n-1)!}{k!(n-k)!}.$$

Since

$$n(n-1)! = n!,$$

we obtain

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{n!}{k!(n-k)!} = \binom{n}{k}.$$

Hence,

$$\boxed{\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}}.$$

e. Now consider choosing $k$ objects from a set of $n$ objects. Select one particular object, say $x$.

Every choice of $k$ objects falls into exactly one of two cases:

- The choice contains $x$. Then we must choose the remaining $k-1$ objects from the other $n-1$ objects. There are

$$\binom{n-1}{k-1}$$

such choices.

- The choice does not contain $x$. Then all $k$ objects must be chosen from the other $n-1$ objects. There are

$$\binom{n-1}{k}$$

such choices.

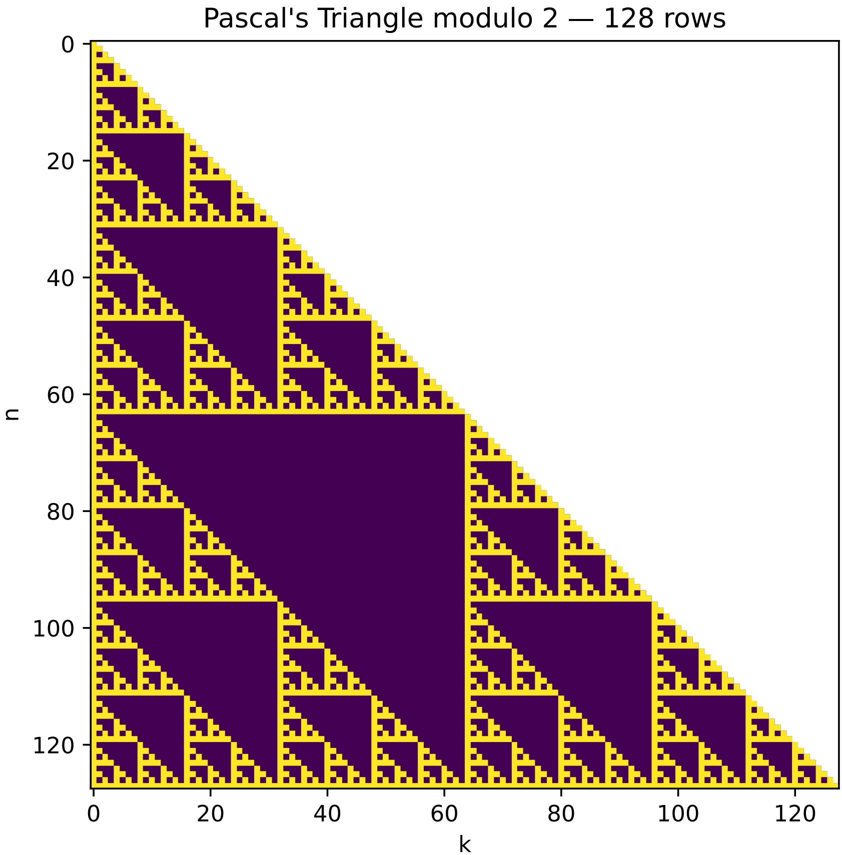
Since these two cases are disjoint and include all possible choices,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

## Solution for Exercise 3.8

```
from sympy import binomial
import numpy as np
import matplotlib.pyplot as plt

nrows = 128

# Build parity matrix: 1 = odd, 0 = even
T = np.zeros((nrows, nrows), dtype=int)

for n in range(nrows):
    for k in range(n + 1):
        T[n, k] = int(binomial(n, k) % 2)

# Plot only the valid triangular region cleanly
masked = np.ma.masked_where(
    np.fromfunction(lambda n, k: k > n, (nrows, nrows)),
    T
)

plt.figure(figsize=(6, 6))
plt.imshow(masked, origin="upper", interpolation="nearest", aspect="equ
plt.title(f"Pascal's Triangle modulo 2 - {nrows} rows")
plt.xlabel("k")
```

```
plt.ylabel("n")
plt.show()
```



# Part III

# Linear algebra

# 4 General vector spaces

A vector space needs vector addition and scalar multiplication, but it need not resemble Euclidean coordinate space.

You may be very familiar with Euclidean space $\mathbb{R}^n$ as a vector space, and quite used to the notion that we have an inner product and a corresponding distance measure on this space. However, linear spaces are much more general. A general vector space does not have an inner product or a norm, just a “linear structure”: Consider for example a general vector space over $\mathbb{R}$: An abstract set $V$ with operations such that one can *add vectors*,

$$z = x + y \in V,$$

and *multiply vectors by constants* $c \in \mathbb{R}$,

$$z = cx \in V$$

where $c \in \mathbb{R}$. The full set of axioms is as follows:

> **Definition: Vector space.** A *vector space over the field* $\mathbb{F}$ is a set $V$ together with a binary *vector addition* $+ : V \times V \to V$ and *scalar multiplication* $\cdot : \mathbb{F} \times V \to V$ such that, for all $x, y, z \in V$ and all $\alpha, \beta \in \mathbb{F}$, the following axioms are true:
>
> 1. There exists a $0 \in V$ such that $0 + x = x$ for all $x \in V$
> 2. $x + (y + z) = (x + y) + z$
> 3. $x + y = y + x$
> 4. There exists $x'$ such that $x + x' = 0$
> 5. $(\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x)$
> 6. $1 \cdot x = x$
> 7. $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$
> 8. $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$

In this exercise, we will train our ability to think abstractly, and also try to get acquainted with general vector spaces.

**Example: Somerset Apple Cake.** Ingredients:

- 170 g butter
- 170 g light brown sugar
- 3 eggs
- 1 tablespoon liquid honey
- 1.5 teaspoons cinnamon
- 0.5 teaspoon cloves
- 0.5 teaspoon nutmeg
- 1 teaspoon baking powder
- 240 g wheat flour
- 700 g apples, diced
- 100 ml apple cider, apple juice, or milk
- 100 g light sultana raisins (optional)

Instructions:

- Cream the room-temperature butter with the sugar until light and fluffy.
- Beat in the eggs and honey.
- Sift the flour, baking powder, and spices. Add them to the mixture and stir until the batter is smooth and lump-free.
- Stir in the milk or apple cider (and the raisins, if you choose to use them; they can be soaked in cider for a couple of hours beforehand, if desired).
- Finally, fold the apple pieces into the batter.
- Pour the batter into a greased round tin with parchment paper at the bottom, or use a deep, round ovenproof dish (24 cm in diameter).

- Bake the cake in the middle of the oven at 160°C for 1.5 hours (check with a cake tester to ensure it's fully baked).
- Serve warm or cold with a dollop of whipped cream, clotted cream, vanilla ice cream, or whatever you like.

## 4.1 Exercises

**Exercise 4.1.**

1. Consider viewing the set $L$ of cake recipe ingredient lists as a vector space. Thus, the components of $L$ are various ingredients and their amounts. What is the dimension of this space? What are the challenges encountered?
2. How would you handle recipe lists with different units, such as pints and liters?
3. Consider the vector space $L$ as a vector space of functions on the form $f : S \to \mathbb{R}$, where $S$ is a set. Which set would $S$ be?
4. Define addition of two recipe ingredient lists. Does the sum define a new cake recipe? What is missing?
5. Consider the scaling of a recipe list to produce a new ingredient list. What are the units of the scaling factor?
6. Not every element in the constructed vector space is a valid recipe list. Which lists must be discarded?
7. Let us turn to the recipe steps. At least some of the steps can be interpreted as *linear operators* on the space of ingredients. For example, creaming butter and sugar transforms these two ingredients to an ingredient named "creamed butter and sugar". Can all steps be treated as linear transformations?

## 4.2 Solutions

### Solution to Exercise 4.1

**For part 1.**

The space has dimension equal to the number of possible ingredients. If this is idealized as infinite, then the space is infinite dimensional. One must consider that each ingredient has a unit attached. Thus, “1 egg” can be a different ingredient than “100 g egg” or “1 lb egg”.

**For part 2.**

It is simplest to standardize the quantities in SI units. Another model could retain several unit systems and convert between cups, deciliters, and grams where a conversion is meaningful.

**For part 3.**

The set of possible (ingredient, unit) pairs, for example.

**For part 5.**

Dimensionless.

**For part 6.**

Lists with negative entries.

# 5 Vectors and matrices

These exercises connect bases and vector decomposition with matrix multiplication and linear systems.

## 5.1 Exercises

### 5.1.1 Basic vectors

**Exercise 5.1.** We work in the space $V = \mathbb{R}^2$, plane vectors. We define two vectors,

$$\mathbf{b}_1 = [1, 2]^T, \quad \mathbf{b}_2 = [-1, 1]^T.$$

The superscript $T$ denotes the transpose, so these are column vectors. Show by elementary means that the two vectors are linearly independent. Why do they form a basis for $V$? Make a 2D drawing or plot of them as arrows from the origin.

**Exercise 5.2.** Let $\mathbf{v} = [2, 2]^T$. Using Gaussian elimination, compute the coefficients $x_1$ and $x_2$ such that

$$\mathbf{v} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2.$$

Illustrate this decomposition in the graph from the previous point, using the parallelogram rule: The sum of two vectors is obtained by placing the origin of one at the end of the second, or vice versa.

**Exercise 5.3.** A person walks from home. First, they walk to the subway station. This is 1 km north and 500 m east as the crow flies. The subway takes the person to the city centre, which 3 km east and 2 km south of the subway station, as the crow flies. The person then walks 200 m south and 400 m west to their office.

What are the coordinates of the office, relative to the person's home?

What is the distance to home, as the crow flies?

**Exercise 5.4.** A boat is sailing on the ocean, at a constant speed 20 km/h relative to the water surface, in the northeast direction. However, the water surface is moving too, at a constant velocity. After two hours, the boat is located 20 km north and 10 km east of the starting point. What is the speed and direction of ocean drift?

**Exercise 5.5.** (Robert Beezer) A three-digit number has two properties. The tens-digit and the ones-digit add up to 5. If the number is written with the digits in the reverse order, and then subtracted from the original number, the result is 792. Use a system of equations to find all of the three-digit numbers with these properties.

### 5.1.2 More vectors and matrices

These exercises are to get practice with computing matrix-matrix products, and to get an intuitive feel for them.

**Exercise 5.6.** Let

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & -1 & 0 \\ 8 & 1 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & -1 \\ 5 & 7 \\ 0 & 7 \end{bmatrix}$$

Which matrix products between pairs of these matrices are defined? Compute the products.

**Exercise 5.7.** Let $A$ be the $n \times n$ matrix defined by $A_{i,i+1} = i^2$ for $1 \leq i < n$, and zero otherwise. Compute the action of $A$ and $A^T$ on the standard basis for $\mathbb{R}^n$. Compute the action of $A$ and $A^T$ on an arbitrary vector $\mathbf{x} \in \mathbb{R}^n$.

**Exercise 5.8.** Let $A \in M(n, m, \mathbb{F})$, $B \in M(m, o, \mathbb{F})$. Write $B$ in terms of column vectors, $B = [\mathbf{b}_1, \cdots, \mathbf{b}_o]$. Prove that

$$AB = [A\mathbf{b}_1, \cdots, A\mathbf{b}_o],$$

i.e., $A$ acts on each individual column of $B$. Similarly, show that, if we write $A$ in terms of its *row vectors*,

$$A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \cdots \\ \mathbf{a}_n^T \end{bmatrix},$$

then

$$AB = \begin{bmatrix} \mathbf{a}_1^T B \\ \mathbf{a}_2^T B \\ \dots \\ \mathbf{a}_n^T B \end{bmatrix}.$$

**Exercise 5.9.** Show that each column of $C = AB$ is a linear combination of the columns of $A$ with coefficients determined by $B$,

$$\mathbf{c}_i = \sum_j \mathbf{a}_j B_{ji}.$$

Show a similar result for the rows of $C$.

## 5.2 Solutions

### Solution to Exercise 5.1

Suppose we write $\mathbf{b}_1 x_1 + \mathbf{b}_2 x_2 = 0$. We must show that $x_1 = x_2 = 0$ is the only solution. By adding the equations, we get $3x1 = 0$, and then by insertion into the first equation $x_2 = 0$. Since we have 2 linearly independent vectors in $\mathbb{R}^2$, these form a basis.

### Solution to Exercise 5.2

The equation gives a linear system

$$\begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

where $b_{ij} = (\mathbf{b}_i)_j$. We get an augmented matrix

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 2 \end{bmatrix}$$

Multiply row 2 by $-1/2$

$$\begin{bmatrix} 1 & -1 & 2 \\ -1 & -1/2 & -1 \end{bmatrix}$$

Add row 1 to row 2:

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & -3/2 & 1 \end{bmatrix}$$

Divide row 2 by $-3/2$:

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2/3 \end{bmatrix}$$

Add row 2 to row 1:

$$\begin{bmatrix} 1 & 0 & 4/3 \\ 0 & 1 & -2/3 \end{bmatrix}$$

This system is equivalent to

$$x_1 = 4/3, \quad x_2 = -2/3,$$

which is the solution.

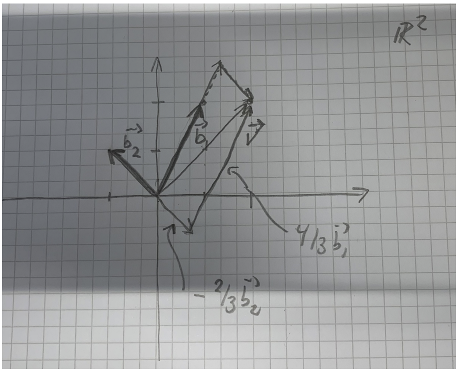



Figure 5.1: image

## Solution to Exercise 5.3

The first point is solved by simple vector addition. The second point is by computing the euclidean norm.

## Solution to Exercise 5.4

Take east and north as the first and second coordinate directions. The boat's velocity relative to the water is

$$\mathbf{v} = \frac{20}{\sqrt{2}}[1,1]^T.$$

Its observed displacement after two hours is $[10, 20]^T$, so

$$2\mathbf{v} + 2\mathbf{u} = [10, 20]^T, \qquad \mathbf{u} = [5 - 10\sqrt{2}, 10 - 10\sqrt{2}]^T \text{ km/h}.$$

The drift speed is $\|\mathbf{u}\|$, and its direction follows from $\operatorname{atan2}(u_2, u_1)$.

## Solution to Exercise 5.5

Let $a$ be the hundreds digit, $b$ the tens digit, and $c$ the ones digit. Then the first condition says that $b + c = 5$. The original number is $100a + 10b + c$, while the reversed number is $100c + 10b + a$. So the second condition is $792 = (100a + 10b + c) - (100c + 10b + a) = 99a - 99c$. So we arrive at the system of equations $b + c = 5$, $99a - 99c = 792$. By elementary row operations, we arrive at the equivalent system $a - c = 8$, $b + c = 5$. We can vary $c$ and obtain infinitely many solutions. However, $c$ must be a digit, restricting us to ten values $(0 - 9)$. Furthermore, if $c > 1$, then the first equation forces $a > 9$, an impossibility. Setting $c = 0$, yields 850 as a solution, and setting $c = 1$ yields 941 as another solution.

## Solution to Exercise 5.6

*The source does not supply a solution.*

## Solution to Exercise 5.7

*The source does not supply a solution.*

## Solution to Exercise 5.8

*The source does not supply a solution.*

### Solution to Exercise 5.9

*The source does not supply a solution.*

# 6 The complex numbers as a real vector space

Identifying $x + \mathrm{i}y$ with $(x, y)$ turns the complex plane into a two-dimensional real vector space.

The geometrical interpretation of the $\mathbb{C}$ as the plane $\mathbb{R}^2$ indicates that $\mathbb{C}$ can be viewed as a two-dimensional real vector space. Indeed, addition and multiplication with *real* scalars are compatible with the axioms for Euclidean space $\mathbb{R}^2$.

## 6.1 Exercises

**Exercise 6.1.**

1. Show that $\mathbb{C}$ regarded can be regarded as the Euclidean plane: Check axioms for Euclidean space and check that vector addition and multiplication with real scalars are the same in the two spaces. Check also that the Euclidean norm is the modulus of the complex number. What are the complex numbers that correspond to the standard basis in $\mathbb{R}^2$? Conclude that $\mathbb{C}$ can be regarded as a *real* Euclidean vector space of dimension 2.
2. Show that multiplication with a *complex* number $z$ is a linear operator on $\mathbb{R}^2$, and find its matrix in the standard basis. What is the matrix of multiplication with i?
3. Show that multiplication with a complex number of modulus 1 is a rotation in $\mathbb{R}^2$.
4. Show that the map $z \mapsto \bar{z}$ is a linear transformation in $\mathbb{R}^2$. What kind of linear transformation is this? Is it a linear transformation in $\mathbb{C}$?

## 6.2 Solutions

### 6.2.1 Tutor note: The complex numbers as a real vector space

This exercise is not essential, so can be skipped for most students.

#### Solution to Exercise 6.1

**Source solution 1.**

The standard basis of $\mathbb{R}^2$ corresponds to 1 and i.

**Source solution 2.**

```
$$(a + \mathrm{i}b)(x + \mathrm{i}y) = (ax - by) + \mathrm{i}(ay + bx)$
    a & -b \\ b & a
\end{bmatrix}$$
```

**Source solution 3.**

Strictly speaking, Euler's formula $e^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ has not been introduced, so "cannot" be used here. But a complex number of modulus 1 forms an angle $\theta$ with the $x$-axis. Since complex multiplication now will not change the length of $z$, only add the angles, this is a rotation.

# 7 Gaussian elimination

Elementary row operations solve linear systems, compute inverses, and expose the rank and consistency of a system.

## 7.1 Exercises

**Exercise 7.1.** Compute the overlap matrix $S$ of the basis defined in the vectors and matrices chapter. Use Gaussian elimination to compute $S^{-1}$.

**Exercise 7.2.** Write down the dual basis of $B = \{\mathbf{b}_1, \mathbf{b}_2\}$.

**Exercise 7.3.** (Robert Beezer) Consider the following system of equations:

$$\begin{aligned}
2x_1 - 3x_2 + x_3 + 7x_4 &= 14 \\
2x_1 + 8x_2 - 4x_3 + 5x_4 &= -1 \\
x_1 + 3x_2 - 3x_3 &= 4 \\
-5x_1 + 2x_2 + 3x_3 + 4x_4 &= -19
\end{aligned}$$

Use Gaussian elimination to find all possible solutions. Write the solution set using set notation.

**Exercise 7.4.** (Robert Beezer) Find all possible solutions of the linear system

$$\begin{aligned}
3x_1 + 4x_2 - x_3 + 2x_4 &= 6 \\
x_1 - 2x_2 + 3x_3 + x_4 &= 2 \\
10x_2 - 10x_3 - x_4 &= 1
\end{aligned}$$

Write down the solution set using set notation.

**Exercise 7.5.** (Robert Beezer) Find all possible solutions of the linear system

$$\begin{aligned} 2x_1 + 4x_2 + 5x_3 + 7x_4 &= -26 \\ x_1 + 2x_2 + x_3 - x_4 &= -4 \\ -2x_1 - 4x_2 + x_3 + 11x_4 &= -10 \end{aligned}$$

Write down the solution set using set notation.

**Exercise 7.6.** Let $\mathbf{b}_i$, $i = 1, 2, \cdots, n$ be a basis for $\mathbb{R}^n$. Let

$$B = [\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_n]$$

be the *basis matrix*, whose columns are precisely the $\mathbf{b}_i$. Explain that if $\mathbf{y} = x_1\mathbf{b}_1 + \cdots + x_n\mathbf{b}_n$, then

$$B\mathbf{x} = \mathbf{y}.$$

Solve for $\mathbf{x}$ using $B$ in symbols.

**Exercise 7.7.** Let $n$ linearly independent vectors $B = \{\mathbf{b}_1, \cdots, \mathbf{b}_n\}$ of $\mathbb{F}^n$ be given. Write down a formula for the dual basis in terms of the overlap matrix.

## 7.2 Solutions

### 7.2.1 Tutor note: Elementary row operations and Gaussian elimination

Consider the system

$$B\mathbf{x} = \mathbf{v}.$$

Here, $B = [\mathbf{b}_1, \mathbf{b}_2]$ is the matrix that has the basis vectors as columns. This system is precisely what we solved above in the two-dimensional case.

We multiply by $B^T$ to get

$$S\mathbf{x} = B^T\mathbf{v}.$$

We invert $S$ to get

$$\mathbf{x} = S^{-1}B^T\mathbf{v}.$$

## Solution to Exercise 7.1

The overlap matrix has entries $S_{ij} = \mathbf{b}_i^T \mathbf{b}_j$, so

$$S = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix}.$$

Row reduction of the augmented matrix gives

$$\left(\begin{array}{cc|cc} 5 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{cc|cc} 1 & 0 & 2/9 & -1/9 \\ 0 & 1 & -1/9 & 5/9 \end{array}\right).$$

Therefore

$$S^{-1} = \frac{1}{9}\begin{pmatrix} 2 & -1 \\ -1 & 5 \end{pmatrix}.$$

## Solution to Exercise 7.2

*The source does not supply a solution.*

## Solution to Exercise 7.3

Using elementary operations, we get the augmented matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & -4 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

The solution is unique: $x_1 = 1$, $x_2 = -3$, $x_3 = -4$, and $x_4 = 1$. Thus

$$S = \{[1, -3, -4, 1]^T\}.$$

### Solution to Exercise 7.4

By elementary row operations, we obtain the augmented matrix:

$$\begin{bmatrix} 1 & 0 & 1 & 4/5 & 0 \\ 0 & 1 & -1 & -1/10 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The last row represents the equation $0 = 1$, so there cannot be a solution to the system. $S = \{\} = \emptyset$.

### Solution to Exercise 7.5

The augmented matrix reduces to

$$\begin{pmatrix} 1 & 2 & 0 & -4 & 2 \\ 0 & 0 & 1 & 3 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Hence $x_2$ and $x_4$ are free, while

$$x_1 = -2x_2 + 4x_4 + 2, \qquad x_3 = -3x_4 - 6.$$

The solution set is

$$S = \{[-2x_2 + 4x_4 + 2, x_2, -3x_4 - 6, x_4]^T \mid x_2, x_4 \in \mathbb{F}\}.$$

### Solution to Exercise 7.6

We know that the overlap matrix should be key here. So we multiply with $B^T$ from the left to get $B^T B\mathbf{x} = B^T\mathbf{y}$. Thus,

$$\mathbf{x} = S^{-1}B^T\mathbf{y}.$$

We know that the overlap matrix $S = B^T B$ is invertible, since $B$ is a basis matrix.

## Solution to Exercise 7.7

The dual basis $\tilde{B} = [\tilde{\mathbf{b}}_1, \cdots \tilde{\mathbf{b}}_n]$ is such that $\langle \tilde{\mathbf{b}}_i, \mathbf{b}_j \rangle = \delta_{ij}$. We obtain the equation

$$\tilde{B}^H B = I.$$

Thus,

$$\tilde{B} = (B^{-1})^H.$$

# 8 Bra-ket notation

Bra-ket notation represents vectors, inner products, operators, and basis expansions in a form used throughout quantum mechanics.

## 8.1 Exercises

**Exercise 8.1.** Let $B = \{b_1, b_2, \cdots, b_n\}$ be a basis for a finite-dimensional Hilbert space $V$, and let $\{|i\rangle\}$ be the standard basis for $\mathbb{C}^n$. Consider the operator

$$\hat{B} = \sum_{i=1}^{n} |b_i\rangle \langle i| .$$

What are the domain and range of $\hat{B}$?

**Exercise 8.2.** Let $|x\rangle \in \mathbb{C}^n$, and consider

$$|\psi\rangle = \hat{B} |x\rangle .$$

Can you interpret this expression in terms of bases and expansion coefficients?

**Exercise 8.3.** Show that

$$\hat{S} = \sum_{ij} |i\rangle \langle b_i|b_j\rangle \langle j| .$$

This is the overlap matrix expressed as an operator. Find an expression for the basis coefficients $x_i$ in terms of the basis alone.

## 8.2 Solutions

### 8.2.1 Tutor note: Bra-ket notation

I think it is important that the students try to relate bra-ket notation to more standard linear algebra notation.

### Solution to Exercise 8.1

The domain is $\mathbb{C}^n$, and the range is $V$. It takes vectors in $\mathbb{C}^n$ and moves them to $V$.

### Solution to Exercise 8.2

The components $x_i = \langle i|x\rangle$ are the expansion coefficients of $|\psi\rangle$ in the basis $B$.

### Solution to Exercise 8.3

$|x\rangle = \hat{S}^{-1}\hat{B}^{\dagger}\,|\psi\rangle$.

# 9 Eigenvalues and diagonalization

These exercises develop the spectral theorem and apply determinants to finite-dimensional eigenvalue problems.

## 9.1 Exercises

### 9.1.1 Eigenvalue decomposition

Of great importance to quantum chemistry is the time-independent Schrödinger equation: Given the Hamiltonian operator $\hat{H}$, which is usually Hermitian, find a nonzero $|\psi\rangle$, called an eigenvector, and a real number $E$, called an eigenvalue, such that

$$\hat{H}\,|\psi\rangle = E\,|\psi\rangle\,.$$

Choosing a particular orthonormal basis $\{|\phi_i\rangle\}$ for Hilbert space, a matrix eigenvalue problem arises,

$$A\mathbf{u} = E\mathbf{u}, \quad u_i = \langle\phi_i|\psi\rangle\,, \quad A_{ij} = \left\langle\phi_i|\hat{H}|\phi_j\right\rangle.$$

We prove the spectral theorem for normal operators over finite dimensional Hilbert spaces. The spectral theorem states that one can actually find a *basis* of eigenvectors for normal operators. The normal operators include the Hermitian operators.

In this section, we let $M(n) = M(n, n, \mathbb{C}) = \mathbb{C}^{n\times n}$ be the set of complex square matrices of dimension $n$.

Two matrices $A$ and $B$ are said to be *similar* if there is an invertible matrix $U$ such that $A = UBU^{-1}$. If $U$ is unitary, then $A$ and $B$ are *unitarily similar*. ($U$ is unitary if and only if $U^H = U^{-1}$.)

**Exercise 9.1.**

1. What does it mean that $A \in M(n)$ is normal?
2. Prove that a diagonal matrix $D \in M(n)$ is normal. (A matrix $A$ is diagonal if and only if $A_{ij} = 0$ whenever $i \neq j$.)
3. Prove that $U$ is unitary if and only if its columns are orthonormal.
4. Prove that if $A$ is normal and $B$ is unitarily similar to $A$, then $B$ is normal.
5. Prove that if $A$ is unitarily equivalent to a diagonal matrix, then $A$ is normal.

The next part requires *Schur's Lemma*:

> **Lemma: Schur's Lemma.** Let $A \in M(n)$. Then $A$ is unitarily equivalent to an upper triangular matrix. ($R$ is upper triangular if $R_{ij} = 0$ whenever $i > j$.)

The proof of Shur's Lemma can be found in, e.g., Fraleigh and Beauregard.

1. Prove that every normal matrix $A$ is unitarily equivalent to a normal upper-triangular matrix $B$. (Use Schur's Lemma and exercise that normality is preserved by unitary similarity.)
2. Prove that a normal upper triangular matrix $B$ must be diagonal. Hint: Let $C = B^H B = BB^H$, and compute $C_{11}$ to show that $B_{1j} = 0$ if $j > 1$. Continue to $C_{22}$, and all the way up to $C_{nn}$.

We have now proved:

> **Theorem: Spectral theorem for normal matrices.** Suppose $A \in M(n)$ is normal, i.e., $AA^H = A^H A$. Then there is a unitary matrix $U \in M(n)$ and a diagonal matrix $D \in M(n)$ such that
>
> $$A = UDU^H. \quad \text{spectral decomposition}$$

1. Let $\mathbf{u}_i = U_{:,i}$ be the $i$th column for $U$. Explain why these vectors form a basis for $\mathbb{C}^n$.
2. Show that

   $$A = \sum_{i=1}^{n} d_i \mathbf{u}_i \mathbf{u}_i^H, \quad d_i = D_{ii}. \quad \text{spectral decomposition}$$

   Write this formula also using bra-ket notation.

3. What does $A$ do with the vector $\mathbf{u}_k$ for some $k$? What does $A$ do to a linear combination of the basis vectors?

A special kind of normal matrix is a Hermitian matrix, for which

$$A^H = A.$$

1. Show that a Hermitian matrix is unitarily equivalent to a diagonal matrix with *real* numbers on the diagonal, i.e., $A = UDU^H$ with $D_{ii} \in \mathbb{R}$.

### 9.1.2 Diagonalization of a matrix

A matrix $A$ has an eigenvalue $E$ if and only if there is a nonzero solution to the linear system

$$(A - EI)\mathbf{x} = 0.$$

Thus, the matrix $A - EI$ must be singular, i.e., not invertible. The eigenvector $\mathbf{x}$ is then a vector in the null space of this matrix.

A sufficient and necessary criterion for invertibility of a matrix $B$ is that the *determinant* is nonzero. The definition of the determinant can be stated as follows:

> **Definition: Determinant.** Let $A \in M(n)$. The determinant, written $\det(A) \in \mathbb{C}$ is defined by
>
> $$\det(A) = \sum_{p \in S_n} (-1)^{|p|} A_{1,p(1)} A_{2,p(2)} \cdots A_{n,p(n)}.$$
>
> Here, $S_N$ is the set of *permutations* of the numbers $\{1, 2, \cdots, n\}$. A permutation is by definition any bijective map.

Each permutation $p \in S_n$ is a rearrangement is thus completely specified by the new order of the numbers $\{1, 2, \cdots, n\}$, i.e., $(p(1), p(2), \cdots, p(n))$. For example, $(1, 2, 3, 4, 5, 6) \in S_6$ is the *identity permutation* that leaves any number alone. The permutation $(3, 2, 1, 4, 6, 5)$ makes 1 and 3 switch place, and 5 and 6.

Any permutation can be written as a product (composition) of *transpositions*, i.e., exchanges of exactly two elements. Thus, $(3, 1, 2) = (1, 2)(2, 3)$. Here, the two-element notation indicates which elements are switched. Every decomposition of a given permutation uses either an even number of transpositions or an odd number, denoted $|p|$ modulo 2. The *sign* of a permutation $p$ is $(-1)^{|p|}$.

There are $n!$ unique permutations of an $n$-element set.

**Exercise 9.2.** Write out the set of permutations $S_1$, $S_2$ and $S_3$.

**Exercise 9.3.** Compute the sign of the permutations of the previous exercise.

**Exercise 9.4.** Compute explicit expressions for the determinants of matrices of dimension 1, 2, and 3.

**Exercise 9.5.** (Extra.) In Python, the module `itertools` contains an iterator class `permutations` that allow efficient looping over permutations. Can you write a function that computes the sign of a permutation? Use the internet for efficiency.

**Exercise 9.6.** Compute explicit expressions for the determinants of matrices of dimension up to 3. By equating the determinant of $A - EI$ with zero, a polynomial equation in $E$ is obtained. Find the solutions in the case $n = 2$.

**Exercise 9.7.** Consider the matrix

$$A = \begin{bmatrix} -1 & z \\ z & 1 \end{bmatrix}.$$

For which $z \in \mathbb{C}$ is $A$ Hermitian? Normal? Compute the eigenvalues as function of $z$. Also find the eigenvectors.

**Exercise 9.8.** For which values of $z$ does there exist a basis of eigenvectors?

## 9.2 Solutions

### 9.2.1 Tutor note: Eigenvalue decomposition

**Tutor note:** This exercise may be considered for the especially interested student. It requires some mathematical argument skill. I would not consider it *hard*, though, but perhaps not focus on this exercise to begin with. I will try to make a solution for this in time.

#### Solution to Exercise 9.1

*The source does not supply a solution.*

### 9.2.2 Tutor note: Diagonalization of a matrix

**Tutor note:** This exercise, I find quite important for understanding eigenvalue problems. I therefore recommend it. The students could also use computations in Python or similar if they find the analytic part a little hard.

I think this is one of th exercises that the tutors should spend some time with.

#### Solution to Exercise 9.2

*The source does not supply a solution.*

#### Solution to Exercise 9.3

*The source does not supply a solution.*

#### Solution to Exercise 9.4

*The source does not supply a solution.*

#### Solution to Exercise 9.5

*The source does not supply a solution.*

#### Solution to Exercise 9.6

*The source does not supply a solution.*

#### Solution to Exercise 9.7

*The source does not supply a solution.*

### Solution to Exercise 9.8

*The source does not supply a solution.*

# 10 Gram–Schmidt orthogonalization

Orthogonal projection leads to the Gram–Schmidt construction and the QR factorization of a full-rank matrix.

Let $V$ be a vector space over $\mathbb{F}$ with $\dim(V) = n < +\infty$. Recall that every finite-dimensional vector space has a basis. In the lecture notes, it is claimed that every finite dimensional Hilbert space has a basis of orthonormal vectors! Recall, that in an orthonormal basis, computations are usually much simpler than in nonorthogonal bases.

In this exercise, we show that any set of linearly independent vectors can be orthonormalized. We work in the space $\mathbb{F}^n$ for simplicity. That is, let $B = [\mathbf{b}_1, \cdots, \mathbf{b}_m] \subset \mathbb{F}^n$ be a set of $m \leq n$ linearly independent vectors, spanning a subspace $V = \text{span}(B)$. We denote the vector set by its basis matrix $B$. Equivalently, $B$ is an arbitrary matrix in $M(n, m, \mathbb{F})$ of full rank $m$, and the vectors $\mathbf{b}_i$ are the columns of $B$.

We will find a set of orthonormal vectors $U = [\mathbf{u}_1, \cdots, \mathbf{u}_m]$, again denoted by a matrix in $M(n, m, \mathbb{F})$, such that $\text{span}(B) = \text{span}(U)$. That is, the two sets of vectors are bases for the *same* subspace, but one of them is an orthonormal basis.

We first define the notion of orthogonal projection onto the span of a single vector: Let $\mathbf{y} \in \mathbb{F}^n$ be nonzero, $Y = \text{span}(\mathbf{y})$, and let $\mathbf{x} \in \mathbb{F}^n$.

## 10.1 Exercises

**Exercise 10.1.**

1. Show that the vector $\mathbf{x}_Y$ given by

$$\mathbf{x}_Y = \mathbf{y}\frac{\langle \mathbf{y}, \mathbf{x}\rangle}{\langle \mathbf{y}, \mathbf{y}\rangle}$$

is the *unique* vector in $Y$, such that

$$\forall \mathbf{y}' \in Y, \quad \|\mathbf{x} - \mathbf{x}_Y\| \leq \|\mathbf{x} - \mathbf{y}'\|.$$

Hint: Find a function $f : \mathbb{R} \to \mathbb{R}$ which you can study, by using the definition of $Y = \text{span}(\mathbf{y})$.

2. Show that

$$\mathbf{x} = \mathbf{x}_Y + \mathbf{x}_{Y\perp},$$

where

$$\langle \mathbf{x}_{Y\perp} | \mathbf{x}_Y \rangle = 0.$$

Illustrate with a picture of the situation in $\mathbb{R}^2$.

The vector $\mathbf{x}_Y$ is called the *orthogonal projection of* $\mathbf{x}$ *onto* $Y$ *or* $\mathbf{y}$.

1. Explain that the orthogonal projection is given by acting with an operator,

$$\mathbf{x}_Y = P_{\mathbf{y}}\mathbf{x}, \quad P_{\mathbf{y}} = \mathbf{y} \langle \mathbf{y}, \mathbf{y} \rangle^{-1} \mathbf{y}^H.$$

The operator $P_{\mathbf{y}}$ is called the *orthogonal projection operator onto* $Y$ = span($\mathbf{y}$).

1. Show that $P^2 = P$ and that $P^\dagger = P$.

The Gram–Schmidt orthogonalization proceeds recursively:

1. $\mathbf{u}_1 = \mathbf{b}_1 \langle \mathbf{b}_1, \mathbf{b}_1 \rangle^{-1/2}$ (normalization)
2. $\mathbf{u}_2' = \mathbf{b}_2 - P_{\mathbf{u}_1}\mathbf{b}_2$ (removal of projection), and $\mathbf{u}_2 = \mathbf{u}_2' \langle \mathbf{u}_2', \mathbf{u}_2' \rangle^{-1/2}$ (normalization)
3. $\mathbf{u}_3' = \mathbf{b}_3 - P_{\mathbf{u}_1}\mathbf{b}_3 - P_{\mathbf{u}_2}\mathbf{b}_3$ (removal of projection), and $\mathbf{u}_3 = \mathbf{u}_3' \langle \mathbf{u}_3', \mathbf{u}_3' \rangle^{-1/2}$ (normalization)
4. Continue recursively.
5. $\mathbf{u}_m' = \mathbf{b}_m - \sum_{i=1}^{m-1} P_{\mathbf{u}_i}\mathbf{b}_m$ (removal of projection), and $\mathbf{u}_m = \mathbf{u}_m' \langle \mathbf{u}_m', \mathbf{u}_m' \rangle^{-1/2}$ (normalization)

We see that each $\mathbf{u}_i$ is given by projecting away from $\mathbf{b}_i$ the components of all the *previously generated* $\mathbf{u}_j$, $j < i$.

1. Show that all the generated vectors $\mathbf{u}_i$ are orthonormal.
2. Show that a set of orthonormal vectors are linearly independent.

3. Show that the span of the first $k$ original vectors is the same as the span of the first $k$ generated vectors. Show that

$$B = UR,$$

where $R \in M(m, m, \mathbb{F})$ is *upper triangular*, i.e., $R_{ij} = 0$ whenever $i > j$. This is called the QR decomposition of the operator $B$. (The "Q" is just our $U$.)

4. We have shown the existence of the QR decomposition when $B$ had full rank. Can you show the existence of the decomposition when $B$ does *not* have full rank? It always exists. Hint: What happens when $B$ does not have full rank?

We now turn to a computational exercise:

1. Write a Python function to compute the QR decomposition of a matrix with full rank. In SciPy, the function `scipy.linalg.qr` is an advanced implementation with pivoting. To compare with your implementation, use the call

```
Q,R = qr(A,mode='economic',pivoting=False)
```

With `pivoting=False`, the function computes essentially the same as our presented algorithm. (Turning on pivoting stabilizes the algorithm, rearranging the vectors before orthogonalizing.)

2. Let $n \geq m > 0$ be integers, and define the matrix $A \in \mathbb{R}^{n\times m}$ by

$$A_{ij} = \left(\frac{i-1}{n-1}\right)^{j-1}.$$

This the matrix of the $m$ first monomials $x^{j-1}$ evaluated at an equidistant grid of $n$ points in the interval $[0, 1]$. Test your QR implementation on $A$ for various $n, m$, say $m = 10$ and $m = 200$. In particular, compute the difference between your factorization and $A$, and compare also with the NumPy implementation.

## 10.2 Solutions

### 10.2.1 Tutor note: Gram–Schmidt orthogonalization

**Tutor note:** I think Gram–Schmidt is important, but maybe there is not enough time in the exercise session.

This exercise may be hard for students who have little experience with proofs. Tutors should prepare the projection and span arguments.

### Solution to Exercise 10.1

**For the rank-deficient QR question.**

Sketch: When $B$ has rank $m' < m$, remove $m - m'$ dependent columns and factor the reduced matrix. Extend the resulting orthonormal set with $m - m'$ vectors orthogonal to those already generated, and add the corresponding rows and columns to $R$ so that $B = UR$ still holds.

# 11 The singular-value decomposition

The singular-value decomposition gives orthogonal matrix expansions and low-rank approximations, including image compression.

The SVD is a powerful linear algebra tool. In quantum chemistry it is used to compress electron-repulsion integrals, for example.

In this small exercise, we demonstrate its power applied to *image compression*. For this exercise, a good grayscale image is handy. You can find a nice picture of a parrot in Figure 11.1.

Recall first the *Hilbert–Schmidt* inner product on the set $M(n, m, \mathbb{F})$ of matrices:

$$\langle X, Y \rangle_{\mathrm{HS}} = \mathrm{Tr}(X^H Y)$$

The *trace* $\mathrm{Tr}(A)$ is the sum of the diagonal elements of $A$.

Recall also the SVD: For $A \in M(n, m, \mathbb{F})$ and $k = \min(n, m)$, there are unitary matrices $U \in M(n)$ and $V \in M(m)$ and singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$ such that

$$A = U\Sigma V^H = \sum_{i=1}^{k} \mathbf{u}_i \sigma_i \mathbf{v}_i^H.$$

Here $\Sigma \in M(n, m, \mathbb{R})$ has the singular values in its upper $k \times k$ diagonal block and zeros elsewhere. Removing unused columns gives the economical SVD

$$A = U_{:,:k} \Sigma_{:k,:k} V_{:,:k}^H.$$

Figure 11.1: A parrot, https://www.dropbox.com/s/8r4svs6uuhgwawt/parrot.png

## 11.1 Exercises

**Exercise 11.1.**

1. How is the Hilbert–Schmidt inner product related to the Euclidean inner product and the standard Euclidean vector space?
2. Show that $\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB)$ (cyclic invariance).
3. Let $\mathbf{u}_i$ and $\mathbf{v}_i$ be the left and right singular vectors of a matrix $A$. Show that the matrices $X^{ij} = \mathbf{u}_i\mathbf{v}_j^H$ form an orthonormal set in the Hilbert–Schmidt inner product.
4. Conclude that the SVD is an expansion of a matrix in an orthonormal basis.

The *truncated* SVD

$$A^{(m)} = \sum_{i=1}^{m} \mathbf{u}_i\sigma_i\mathbf{v}_i^H$$

is the best approximation to $A$ of rank at most $m$ in the Hilbert–Schmidt norm:

$$A^{(m)} = \underset{\mathrm{rank}(B)\leq m}{\arg\min} \|B - A\|_{\mathrm{HS}}.$$

1. Write a Python program/notebook that reads an image file and converts it to a matrix $A \in M(n, m, \mathbb{R})$. The module `imageio` can be useful.
2. Continue writing the program, so it computes the truncated and full SVD of $A$. You can use, say, `scipy.linalg.svd` to find the full SVD.
3. Plot the singular values. Discuss.
4. Show that the error in the Hilbert–Schmidt norm is

$$\|A^{(m)} - A\|_{\mathrm{HS}} = \sqrt{\sum_{i=m+1}^{k} \sigma_i^2}.$$

5. Make visualizations of the truncated SVD of the image $A$, with various $m$. Compute the relative error $\|A^{(m)} - A\|_{\mathrm{HS}}/\|A\|_{\mathrm{HS}}$ for each $m$ you consider. Discuss the image quality.

## 11.2 Solutions

### 11.2.1 Tutor note: The singular value decomposition as a compression tool

**Tutor note:** This exercise combines computations with accessible analysis. It will probably take most of an hour, depending on the student.

#### Solution to Exercise 11.1

**For the orthonormal-matrix question.**

$$\left\langle X^{ij}, X^{i'j'} \right\rangle_{\mathrm{HS}} = \mathrm{Tr}(\mathbf{v}_j \mathbf{u}_i^H \mathbf{u}_{i'} \mathbf{v}_{j'}^H) = \langle \mathbf{u}_i, \mathbf{u}_{i'} \rangle \left\langle \mathbf{v}_j, \mathbf{v}_{j'} \right\rangle = \delta_{ii'} \delta_{jj'}.$$

# 12 Polynomial vector spaces

Polynomials form finite-dimensional vector spaces on which differentiation, multiplication, and changes of basis act as linear maps.

In this exercise, we study a vector space that is not $\mathbb{F}^n$. We start out without an inner product, but supply this later.

Recall that a polynomial of degree $n$ over $\mathbb{F}$ has the form

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$

with $a_i \in \mathbb{F}$ and $a_n \neq 0$.

## 12.1 Exercises

**Exercise 12.1.** Let $P_n$ be the set of all polynomials of degree less than or equal to $n$. Show that $P_n$ is a vector space of dimension $n + 1$ over $\mathbb{C}$.

**Exercise 12.2.** Let $D : P_n \to P_n$ be the differentiation operator, i.e.,

$$(Dp)(x) = p'(x).$$

Show that $D$ is indeed a linear operator.

**Exercise 12.3.** Does there exist an inverse of $D$?

**Exercise 12.4.** Let $B = \{1, x, x^2, \cdots, x^n\}$ be the basis of *monomials*. Using bra-ket notation, we write the basis operator as

$$\hat{B} = \sum_{i=1}^{n+1} \left|x^{i-1}\right\rangle \langle i| \,,$$

where $|i\rangle$ is a standard basis vector for $\mathbb{F}^{n+1}$. Compute the matrix of $D$ relative to this basis. (Note carefully that we do not have an inner product defined.) That is, find a matrix $T$ such that

$$D\hat{B} = \hat{B}T.$$

We now restrict our attention to the interval $x \in [-1,1]$. That is,

$$V_n = \{p : [-1,1] \to \mathbb{C} \mid p \text{ a polynomial of deg} \leq n\}.$$

The above subexercises did not depend on the domain of the polynomial.

We define an inner product on $V$,

$$\langle p|q\rangle := \int_{-1}^{1} \overline{p(x)}q(x)\,\mathrm{d}x.$$

We will now define the *Legendre polynomials.*

**Exercise 12.5.** Show that $\langle\cdot|\cdot\rangle$ is indeed an inner product by checking the axioms.

**Exercise 12.6.** Compute the overlap matrix $S$ of the monomial basis. Is the basis orthonormal?

**Exercise 12.7.** Formulate the Gram–Schmidt procedure using the bra-ket notation. Explain how the procedure generates an upper triangular matrix $R$ such that

$$\hat{B} = \hat{U}R,$$

where $\hat{U}$ is the basis matrix of the orthonormal set of vectors.

**Exercise 12.8.** Use the Gram–Schmidt decomposition by hand to compute the first three *normalized Legendre polynomials* $|\hat{P}_i\rangle$, $i = 0,1,2$, defined using the Gram–Schmidt orthogonalization of $\{1, x, x^2\}$.

The Legendre polynomials are orthogonal polynomials $P_i : [-1,1] \to \mathbb{R}$ obtained from the monomials by Gram–Schmidt orthogonalization and normalized by $P_i(1) = 1$. They are orthogonal but not normalized. The endpoint normalization gives

$$\int_{-1}^{1} P_i(x)P_j(x)\,\mathrm{d}x = \frac{2}{2i+1}\delta_{ij}.$$

With $P_0(x) = 1$ and $P_{-1} \equiv 0$, they satisfy the recurrence relation

$$(i+1)P_{i+1}(x) = (2i+1)xP_i(x) - iP_{i-1}(x).$$

From this, one can show:

$$P'_{i+1}(x) = \frac{2P_i(x)}{\|P_i\|^2} + \frac{2P_{i-2}(x)}{\|P_{i-2}\|^2} + \frac{2P_{i-4}(x)}{\|P_{i-4}\|^2} + \dots$$

**Exercise 12.9.** Compute the matrix of the differential operator $D$ in the *normalized* Legendre polynomial basis, both by using matrix multiplication involving $X$, $R$, and $R^{-1}$, and also by using the derivative relation above. Compare the two explicitly using pen-and-paper calculations.

**Exercise 12.10.** Repeat the previous exercise with the operator $D^2$.

Orthogonal polynomials define efficient *quadrature rules*. Consider approximating an integral by

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} f(x_i)w_i,$$

where the $w_i$ are *weights* and the $x_i$ are *nodes*. A unique choice of nodes and weights makes the rule exact for every polynomial of degree at most $2n-1$. The nodes are the zeros of $P_n$, and the weights are

$$w_i = \frac{2}{(1-x_i^2)[P_n'(x_i)]^2}.$$

The Wikipedia page on Gauss–Legendre quadrature contains more details.

The so-called Golub–Welsch algorithm is a simple algorithm based on diagonalization of a tridiagonal Hermitian matrix for computing the nodes and weights of Gaussian quadratures, see the Wikipedia page on Gaussian quadrature. The interested student should check it out!

## 12.2 Solutions

### 12.2.1 Tutor note: Space of polynomials

**Tutor note:** Treating functions as vectors is important in quantum chemistry. The recommended sub-exercises should be broadly accessible; the remaining exercises may require tutor preparation.

#### Solution to Exercise 12.1

The sum of two polynomials is a polynomial. A scalar multiple of a polynomial is a polynomial. The polynomials $x^i$ are clearly linearly independent. Since all polynomials are linear combinations of such, the dimension must be equal to the number of coefficients, $n+1$.

## Solution to Exercise 12.2

*The source does not supply a solution.*

## Solution to Exercise 12.3

No inverse exists, since the null space of $D$ is nontrivial (the constant polynomials). The indefinite integral would be a candidate for inverse, but the integral is only determined up to a constant.

## Solution to Exercise 12.4

Since $(x^{i-1})' = (i-1)x^{i-2}$ for $i > 1$, we get

$$T_{i-1,i} = i - 1, \qquad i = 2, \dots, n + 1,$$

and all other matrix elements are zero.

## Solution to Exercise 12.5

*The source does not supply a solution.*

## Solution to Exercise 12.6

$$\int_{-1}^{1} x^i x^j \, \mathrm{d}x = \frac{1}{i+j+1}(1 - (-1)^{i+j+1}).$$

Clearly not an orthonormal basis, since the matrix is not the identity matrix.

## Solution to Exercise 12.7

*The source does not supply a solution.*

## Solution to Exercise 12.8

*The source does not supply a solution.*

### Solution to Exercise 12.9

*The source does not supply a solution.*

### Solution to Exercise 12.10

*The source does not supply a solution.*

# Part IV

# Metric spaces and topology

# 13 Metric spaces

A metric measures distance through symmetry, positivity, and the triangle inequality.

A *metric* encodes the intuition behind measuring *distances* in a set $M$. It is a distance function $d(x, y)$, where $x, y \in M$.

The definition of a metric space is as follows:

> **Definition: Metric.** Let $M$ be a set. A function $d : M \times M \to \mathbb{R}$ is a *metric* if it satisfies the following axioms:
>
> 1. $d(x, y) = d(y, x)$
> 2. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$
> 3. $d(x, y) \leq d(x, z) + d(z, y)$
>
> The pair $(M, d)$ is a *metric space*. If $M$ is a vector space, we say that $(M, d)$ is a *metric vector space*.

In this exercise, we will try to understand why the axioms are as they are, i.e., what makes them encode the concept of a metric.

## 13.1 Exercises

**Exercise 13.1.**

1. Consider two persons $X$ and $Y$ located at $x$ and $y$ in a set $M$. They both measure the distance to the other person. What would happen if axiom a) is violated? Is this reasonable?
2. What would happen if axiom b) is violated? Is this reasonable?
3. A third person $Z$ located at $z$ enters the picture. Consider axiom c). What happens if this axiom is violated? Is this reasonable?

4. Let $M = \mathbb{R}^2$, and let $d$ be Euclidean distance, "as the crow flies". But there are other distance measures as well. Consider for example the "Manhattan distance", given by

$$d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|.$$

This has a standard interpretation in terms of cities with straight streets and taxis driving on these streets. Discuss.

5. Consider a king moving on a chessboard. The king can move one step at a time, either horizontally or vertically, or diagonally. Consider the *minimum number of steps* the king has to move in order to get from one point to another. See Figure 13.1 for an illustration. The metric is

$$d(\mathbf{x}, \mathbf{y}) = \max(|x_1 - y_1|, |x_2 - y_2|).$$

This is also called the "maximum metric". Can you verify the metric axioms? How does the maximum metric compare to the Manhattan metric?

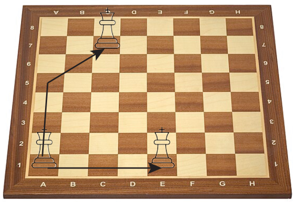


Figure 13.1: Illustration of "chessboard metric": The king can move horizontally, vertically, and diagonally. The chessboard metric is the smallest number of steps the king must move in order to get from one point to another. Here, the king starts at A1, and two paths are shown: The optimal path to C7, and the optimal path to E1. Note that the king will move diagonally as much as he can, because he travels "faster" that way!

## 13.2 Solutions

### Solution to Exercise 13.1

**For part 1.**

The two observers would disagree about the distance between the same pair of points. A directed cost can be asymmetric, but it is not a metric under this definition.

**For part 2.**

Zero distances between non-equal points violate the concept of distances measuring how far apart points are.

**For part 3.**

The triangle inequality embodies an essential feature of Euclidean space: Taking detours should never lead to shorter distances traveled!

**For part 5.**

One possible set of observations is the following: The maximum metric is smaller than the Manhattan metric. On the other hand, the Manhattan metric is smaller than 2 times the maximum metric. This means that open sets in the Manhattan metric will be open sets in the maximum metric, and vice versa, since inside an open ball for each metric, we can always place an open ball of the other metric. Therefore, they create the same topology. A function $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous with respect to one metric iff it is continuous with respect to the other metric. (The same is also true for the Euclidean metric, they are all *equivalent*.)

# 14 Open and closed sets in Euclidean space

Open balls define interior points, open sets, closed sets, boundaries, and closures in Euclidean space.

## 14.1 Exercises

**Exercise 14.1.** Show that any $\epsilon$-ball is open.

**Exercise 14.2.** Show that $A \cup \partial A$ is closed.

**Exercise 14.3.** Show that if $A \subset \mathbb{R}^n$ is closed and a sequence $\mathbf{x}_i \in A$ converges to some $\mathbf{x} \in \mathbb{R}^n$, then $\mathbf{x} \in A$. This gives an equivalent characterization of closed subsets of $\mathbb{R}^n$.

**Exercise 14.4.** Show that the closure of $A$, the smallest closed set $\bar{A}$ that contains $A$, is equal to $A \cup \partial A$.

**Exercise 14.5.** Show that

$$A = \{(x, 0) \mid x \in [-1, 1]\} \subset \mathbb{R}^2$$

is closed.

**Exercise 14.6.** Show that $\mathbb{Q} \subset \mathbb{R}$ is neither open nor closed. Show that the boundary of $\mathbb{Q}$ is $\mathbb{R}$. Show that the interior of $\mathbb{Q}$ is empty. What is the closure of $\mathbb{Q}$?

**Exercise 14.7.** Let

$$A = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x < 1, \quad 0 \leq y < 1\}.$$

Compute the interior of $A$, the boundary of $A$, the closure of $A$.

## 14.2 Solutions

### 14.2.1 Tutor note: Open and closed sets in Euclidean space

**Tutor note:** This exercise really develops the basic skills thinking about open and closed sets, and also mathematical argumentation. Even if the exercises are fairly basic by undergrad maths standards, maybe this style of thinking is difficult for some students, and therefore I think it is worthwhile.

#### Solution to Exercise 14.1

Let $A = B_\epsilon(\mathbf{x}_0)$ and $\mathbf{x} \in A$. Set $r = \|\mathbf{x} - \mathbf{x}_0\| < \epsilon$ and $\delta = \epsilon - r > 0$. If $\mathbf{y} \in B_\delta(\mathbf{x})$, then

$$\|\mathbf{y} - \mathbf{x}_0\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\| < \delta + r = \epsilon.$$

Thus $B_\delta(\mathbf{x}) \subset A$. Every point of $A$ has an open ball contained in $A$, so $A$ is open.

#### Solution to Exercise 14.2

Let $C = (A \cup \partial A)^\complement$. If $\mathbf{x} \in C$, then $\mathbf{x} \notin A$ and $\mathbf{x} \notin \partial A$. Since $\mathbf{x}$ is not a boundary point, some ball $B_\epsilon(\mathbf{x})$ lies wholly in $A$ or wholly in $A^\complement$. It cannot lie in $A$ because it contains $\mathbf{x}$, so it lies in $A^\complement$ and contains no boundary point. Hence $B_\epsilon(\mathbf{x}) \subset C$. The complement $C$ is open, so $A \cup \partial A$ is closed.

#### Solution to Exercise 14.3

*The source does not supply a completed solution.*

## Solution to Exercise 14.4

We have shown that $A \cup \partial A$ is closed. Suppose $A \cup \partial A$ is not the smallest closed set that contains $A$, i.e., $(A \cup \partial A) \setminus \bar{A}$ contains some point $\mathbf{x}$. Since both sets contain $A$, we must have $\mathbf{x} \in \partial A$. From this, we should look for a contradiction, because intuitively, the boundary must be part of $\bar{A}$. For all $\epsilon > 0$, $B_\epsilon(\mathbf{x})$ intersects both $A$ and $A^\complement$. Therefore, there exists a sequence $\mathbf{x}_i \in A$ such that $\mathbf{x}_i \to \mathbf{x} \in \partial A$, but the smallest closed set containing $A$ must contain this $\mathbf{x}$.

## Solution to Exercise 14.5

Consider $A^\complement$, the plane without the interval $[-1, 1]$ on the $x$-axis. For a point $(x, y) \in A^\complement$ with $y \neq 0$, choose any $0 < \epsilon < |y|$. If $y = 0$, then $|x| > 1$, and we may choose any $0 < \epsilon < |x| - 1$. In either case, the $\epsilon$-ball around $(x, y)$ misses $A$. Thus $A^\complement$ is open and $A$ is closed.

## Solution to Exercise 14.6

Let $p/q \in \mathbb{Q}$ be a rational number, and let $]p/q-\epsilon, p/q+\epsilon[$ be an $\epsilon$-"ball" around $p/q$. But we know that between any two real numbers, there are infinitely many rational numbers, and also irrational numbers. Hence, the ball is not contained in $\mathbb{Q}$, so the rational numbers do not form an open set. Hence no rational number has an open ball around it, and hence the interior of $\mathbb{Q}$ is empty. To show that $\mathbb{Q}$ is not closed, consider the openness of the complement, $\mathbb{R} \setminus \mathbb{Q}$, the irrational numbers. The same argument can be applied, the irrationals are not open.

The boundary of $\mathbb{Q}$ is all of $\mathbb{R}$, since any ball around a rational number contains both rational and irrational numbers.

Finally, the closure of $\mathbb{Q}$ is then $\mathbb{R}$.

## Solution to Exercise 14.7

The interior of $A$ is

$$\{(x, y) \in \mathbb{R}^2 \mid 0 < x < 1, \quad 0 < y < 1\}.$$

The boundary of $A$ is the border of the square,

$$\partial A = \{(0,t) \mid 0 \le t \le 1\} \cup \{(1,t) \mid 0 \le t \le 1\} \cup \{(t,0) \mid 0 \le t \le 1\} \cup \{(t,1) \mid 0 \le t \le 1\}$$

The closure of $A$ is $A$ with the border, hence

$$\{(x,y) \in \mathbb{R}^2 \mid 0 \le x \le 1, \quad 0 \le y \le 1\}.$$

# Part V

# Calculus

# Part VI

# Multivariable calculus

# 15 Visualizing functions

Graphs, level sets, and vector fields provide complementary views of scalar- and vector-valued functions.

Some paths and their curves:

## 15.1 Exercises

**Exercise 15.1.** Let $f : \mathbb{R} \to \mathbb{R}^2$ be a path defined by

$$f(t) = [r(t)\cos(4t), r(t)\sin(4t)], \quad r(t) = \exp(-t).$$

Sketch the curve traced by the path for $0 \leq t \leq 2\pi$. You can check your sketch against, say, a Python plot.

**Exercise 15.2.** Let $f : (0, +\infty) \to \mathbb{R}^2$ be a path defined by

$$f(t) = [\ln(t), t].$$

Sketch the curve traced by the path. You can check your sketch against, say, a Python plot.

**Exercise 15.3.** Let $f : [0, \pi] \to \mathbb{R}^2$ be given by $f(t) = [x(t), y(t)]$, with $x(t) = 16\sin(t)^3$ and $y(t) = 13\cos(t) - 5\cos(2t) - 2\cos(3t) - \cos(4t)$. Sketch the curve. Here, it is probably easiest to use Python or a similar tool.

Level curves:

**Exercise 15.4.** Let $f(x, y) = x^2 + y^2$. Sketch the level curves $\{(x, y) \mid f(x, y) = n\}$, for $n = 1$ and $n = 2$, $n = 3$, and $n = 4$.

**Exercise 15.5.** Let $f(x, y) = x + y + 2$. Sketch the level curves where $f(x, y) = 0$, 2 and 4. Can you sketch the graph of $f$?

**Exercise 15.6.** Let $f(x,y) = x^2 - y^2$. The graph is called the *hyperbolic paraboloid*, or a saddle. Sketch the graph. Sketch the level curves $\{(x,y) \mid f(x,y) = n\}$, for $n = 0$ and $n = -1$, $n = 1$. Make sure that you get all "pieces".

**Exercise 15.7.** Write a Python program for sketching the level curves in the hyperbolic-paraboloid exercise above, but use more closely spaced constants. There are tools in Matplotlib that are handy.

**Exercise 15.8.** Adapt your program to draw level curves of $(x,y) \mapsto (x^2 + 3y^2)e^{1-x^2-y^2}$.

Sections and level surfaces:

Consider the hydrogen-atom eigenfunctions in atomic units. They are indexed by $n \geq 1$, $0 \leq \ell < n$, and $-\ell \leq m \leq \ell$. Their separated form is

$$\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi),$$

with

$$R_{n\ell}(r) = N_{n\ell}\rho^{\ell}L_{n-\ell-1}^{2\ell+1}(\rho)e^{-\rho/2}, \qquad \rho = \frac{2r}{n}.$$

Here $N_{n\ell}$ is a normalization constant. Coordinate conversion uses

$$r = \sqrt{x^2+y^2+z^2}, \quad x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta.$$

In particular, the $1s$ and $2p_z$ functions are given by

$$\psi_{1s}(x,y,z) = \frac{1}{\sqrt{\pi}}e^{-r(x,y,z)},$$

$$\psi_{2p_z}(x,y,z) = \frac{1}{4\sqrt{2\pi}}ze^{-r(x,y,z)/2},$$

**Exercise 15.9.** Write a Python program to draw the level *surfaces* of hydrogen orbitals, $\{(x,y,z) \mid \psi(x,y,z) = c\}$. Choose interesting values of $c$. It can be interesting to color code surfaces of opposite positive and negative $c$.

**Exercise 15.10.** Write a program to visualize *sections* of the orbital graphs. For example, fix $y = y_{\text{const}}$ and plot $(x,z) \mapsto \psi(x, y_{\text{const}}, z)$.

## 15.2 Solutions

### 15.2.1 Tutor note: Visualization of functions

**Tutor note:** The first few exercises are not hard, and the plots give quick visual feedback. The later exercises require some straightforward programming. The level-curve exercise comes from Marsden and Tromba.

#### Solution to Exercise 15.1

*The source does not supply a solution.*

#### Solution to Exercise 15.2

*The source does not supply a solution.*

#### Solution to Exercise 15.3

*The source does not supply a solution.*

#### Solution to Exercise 15.4

*The source does not supply a solution.*

#### Solution to Exercise 15.5

*The source does not supply a solution.*

#### Solution to Exercise 15.6

*The source does not supply a solution.*

### Solution to Exercise 15.7

*The source does not supply a solution.*

### Solution to Exercise 15.8

*The source does not supply a solution.*

### Solution to Exercise 15.9

*The source does not supply a solution.*

### Solution to Exercise 15.10

*The source does not supply a solution.*

# 16 Continuity and differentiability

Limits define continuity, while partial derivatives describe local change for functions of several variables.

## 16.1 Exercises

### 16.1.1 Continuity

**Exercise 16.1.** Let $f : \mathbb{R} \to \mathbb{R}$ be given by

$$f(x) = \begin{cases} 1 & x > 0 \\ -1 & x \leq 0 \end{cases}$$

For which $x \in \mathbb{R}$ does the limit

$$\lim_{h \to 0} f(x+h)$$

*not* exist? Why?

**Exercise 16.2.** Show that any polynomial $p : \mathbb{R} \to \mathbb{R}$ is continuous, by using the theorem on properties of continuous functions.

**Exercise 16.3.** (Marsden and Tromba) Consider the function

$$f(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}.$$

This function is defined on $\mathbb{R}^2 \setminus \{(0,0)\}$. Determine whether $f(x, y)$ approaches some value as $(x, y) \to (0,0)$. What is this value?

**Exercise 16.4.** (Marsden and Tromba) Does $\lim_{(x,y)\to(0,0)} x^2/(x^2 + y^2)$ exist? If you want, plotting the function can help.

**Exercise 16.5.** (Marsden and Tromba) Prove that $\lim_{(x,y)\to(0,0)} 2x^2y/(x^2+y^2) = 0$ using an $\epsilon$-$\delta$ argument.

### 16.1.2 Differentiability

**Exercise 16.6.** Compute the partial derivatives of

$$f(x, y) = \frac{xy}{(x^2 + y^2)^{1/2}}.$$

## 16.2 Solutions

### 16.2.1 Tutor note: Continuity

**Tutor note:** The concept of continuity is basic, so I recommend some of these exercises for everyone. I do not think the tutors need much preparation for them.

#### Solution to Exercise 16.1

The limit does not exist at $x = 0$. For every interval $]-\epsilon, \epsilon[$, $f$ maps into $\{-1, 1\}$, which does not approach a single value.

#### Solution to Exercise 16.2

The identity function $x \mapsto x$ is continuous. Since sums and products of continuous functions are continuous, every polynomial is continuous.

#### Solution to Exercise 16.3

Use $\lim_{t\to 0} \sin(t)/t = 1$. Given $\epsilon > 0$, choose $\eta > 0$ such that $0 < |t| < \eta$ implies $|\sin(t)/t - 1| < \epsilon$. Set $\delta = \sqrt{\eta}$. If $0 < \|\mathbf{u}\| < \delta$, then $0 < \|\mathbf{u}\|^2 < \eta$, so

$$\left| \frac{\sin(\|\mathbf{u}\|^2)}{\|\mathbf{u}\|^2} - 1 \right| < \epsilon.$$

Therefore the limit is 1.

### Solution to Exercise 16.4

A hint to give the students:

Consider the line $x = 0$ and the line $y = 0$.

The limit does not exist. Along $y = 0$ the function equals 1, while along $x = 0$ it equals 0.

### Solution to Exercise 16.5

We have

$$|2x^2y/(x^2 + y^2)| \leq |2x^2y/x^2| \leq 2|y|.$$

Thus, given $\epsilon > 0$, choose $\delta = \epsilon/2$. Then $0 < \|(x, y) - (0, 0)\| = \sqrt{x^2 + y^2} < \delta$ implies $|y| < \delta$, and so $|f(x, y) - 0| < 2\delta < \epsilon$.

## 16.2.2 Tutor note: Differentiability

**Tutor note:** The source contains only one exercise on differentiability; more exercises may be added later.

### Solution to Exercise 16.6

For $(x, y) \neq (0, 0)$, write $v = (x^2 + y^2)^{1/2}$. The quotient and chain rules give

$$\frac{\partial f}{\partial x} = \frac{yv - xy(x/v)}{v^2} = \frac{y^3}{(x^2 + y^2)^{3/2}},$$

and, by symmetry,

$$\frac{\partial f}{\partial y} = \frac{x^3}{(x^2 + y^2)^{3/2}}.$$

# Part VII

# Series and matrix functions

# 17 Matrix exponentials and commutators

Power series define matrix exponentials. Differentiating products of exponentials produces nested commutators.

Consider the *matrix-valued function*

$$f(t) = e^{-tA} B e^{tA},$$

where $A$ and $B$ are square matrices. The matrix exponential is defined in terms of the Taylor series,

$$\exp(X) = \mathbf{1} + X + \frac{1}{2!}X^2 + \frac{1}{3!}X^3 + \cdots,$$

which is convergent. It is a fact that if $XY - YX = [X, Y] = 0$ (commuting matrices) then $\exp(X + Y) = \exp(X)\exp(Y)$, by the same proof as for the exponential function over a field $\mathbb{F}$.

## 17.1 Exercises

**Exercise 17.1.** Explain why we may view $f$ as a vector-valued function

**Exercise 17.2.** Show that $t \mapsto e^{tA}$ is differentiable.

**Exercise 17.3.** Show that $f$ is differentiable. Compute its derivatives to arbitrary order

## 17.2 Solutions

### 17.2.1 Tutor note: Taylor polynomials

**Tutor note:** I think it is important in quantum chemistry to be able to view matrices as vectors and as function parameters. On the other hand, some of this exercise may be hard for the student.

The first source item supplied the setup but no question, so it now appears in the chapter introduction.

#### Solution to Exercise 17.1

Identification of $\mathbb{F}^{n\times m}$ with $\mathbb{F}^{nm}$.

#### Solution to Exercise 17.2

Since $tA$ and $hA$ commute,

$$e^{(t+h)A} = e^{tA}e^{hA} = e^{tA}\left(\mathbf{1} + hA + O(h^2)\right).$$

Therefore

$$\lim_{h\to 0} \frac{e^{(t+h)A} - e^{tA}}{h} = e^{tA}A = Ae^{tA}.$$

#### Solution to Exercise 17.3

$f$ is a product of differentiable matrix-valued functions. The product rule gives

$$f'(t) = [f(t), A].$$

Repeated differentiation gives

$$f^{(n)}(t) = \underbrace{[\cdots[[f(t), A], A], \dots, A]}_{n \text{ commutators}}.$$

# Part VIII

# Complex numbers and complex analysis

# 18 Complex arithmetic and geometry

Cartesian and polar forms connect complex arithmetic with plane geometry, rotations, dot products, and cross products.

## 18.1 Exercises

### 18.1.1 Complex numbers

**Exercise 18.1.** Let $z = x + \mathrm{i}y \in \mathbb{C}$, and compute the real and imaginary parts of $z^2$, $z^3$, and $z^4$, as functions of $x$ and $y$.

**Exercise 18.2.** Compute a closed-form expression for the real and imaginary parts of $z^n$.

**Exercise 18.3.** Compute closed-form expression for $z^{-1}$, exhibiting the real and imaginary parts as functions of $x$ and $y$. (Assuming $z \neq 0$). Try to compute closed-form expressions of $z^{-2}$ and $z^n$ as well.

**Exercise 18.4.** A complex number $z = x + \mathrm{i}y$ can be written in polar form,

$$x = r\cos(\theta), \quad y = r\sin(\theta).$$

Write down the rule for $z^n = R\cos\phi + \mathrm{i}R\sin\phi$ using angle $\theta$ and modulus $r$.

**Exercise 18.5.** Consider the polynomial equation $f(z) = z^3 + 1 = 0$. Find all the roots in the complex plane using polar coordinates. Sketch the roots in a coordinate system.

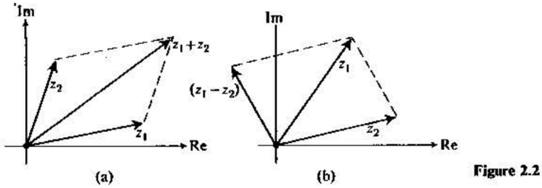



Figure 18.1: The parallelogram rule. (From Butkov.)

### 18.1.2 Basic calculations

**Exercise 18.6.** Let $z = x + \mathrm{i}y \in \mathbb{C}$. Illustrate the addition $w = z + \bar{z}$ in a coordinate system using the parallelogram rule.

**Exercise 18.7.** Recall our earlier exercise, where $\mathbb{C}$ was regarded as the real vector space $\mathbb{R}^2$. Let $z_i = x_i + \mathrm{i}y_i$, and denote the corresponding vectors in $\mathbb{R}^2$ by $\mathbf{v}_i$. Show that

$$\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \Re(\overline{z_1} z_2) = \Re(z_1 \overline{z}_2).$$

**Exercise 18.8.** Another product in $\mathbb{R}^2$ (and indeed $\mathbb{R}^3$) is the *cross product*:

$$\mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{vmatrix},$$

the matrix determinant. Show that

$$\mathbf{v}_1 \times \mathbf{v}_2 = \Im(\overline{z_1} z_2) = -\Im(z_1 \overline{z}_2).$$

**Exercise 18.9.** Show that the function $z \mapsto \mathrm{i}z$ is a counterclockwise rotation by $\pi/2$. Find the function that performs a clockwise rotation by the same angle.

## 18.2 Solutions

### Solution to Exercise 18.1

$$(x + \mathrm{i}y)^2 = x^2 + 2\mathrm{i}xy + \mathrm{i}^2 y^2 = (x^2 - y^2) + 2\mathrm{i}xy.$$

Obtain the next powers by multiplying the preceding result by $z$.

## Solution to Exercise 18.2

The binomial expansion gives

$$(x+\mathrm{i}y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} (\mathrm{i}y)^j.$$

Even powers satisfy $\mathrm{i}^{2k} = (-1)^k$. The even indices between 0 and $n$ are $2j$ for $j = 0, 1, \dots, \lfloor n/2 \rfloor$, while the odd indices are $2j+1$ for $j = 0, 1, \dots, \lfloor (n-1)/2 \rfloor$. Let $K = \lfloor n/2 \rfloor$ and $L = \lfloor (n-1)/2 \rfloor$.

Splitting into even and odd terms gives

$$(x+\mathrm{i}y)^n = \sum_{j=0}^{K} (-1)^j \binom{n}{2j} x^{n-2j} y^{2j} + \mathrm{i} \sum_{j=0}^{L} (-1)^j \binom{n}{2j+1} x^{n-2j-1} y^{2j+1}.$$

## Solution to Exercise 18.3

Multiply by the complex conjugate:

$$z^{-1} = \frac{1}{x+\mathrm{i}y} = \frac{x-\mathrm{i}y}{(x+\mathrm{i}y)(x-\mathrm{i}y)} = \frac{x-\mathrm{i}y}{x^2+y^2}.$$

The formulas from the previous exercise then apply with $y \mapsto -y$ and a prefactor $(x^2+y^2)^{-n}$.

## Solution to Exercise 18.4

The geometric interpretation gives $\phi = n\theta$ (modulus $2\pi$) and $R = r^n$.

## Solution to Exercise 18.5

$-1$ has modulus 1 and angle $\pi$. A root must be such that $r^n = 1$ (which implies $r = 1$), and $3\theta = \pi + 2k\pi$, since the angle wraps around at $2\pi$ when we multiply complex numbers. Thus, $\theta = \pi/3 + 2k\pi/3$. $k = 0$ gives $\pi/3$, $k = 1$ gives $3\pi/3 = \pi$, and $k = 2$ gives $5\pi/3$. Now any larger $k$ wraps around to these three angles. So we have found the roots.

### Solution to Exercise 18.6

*The source does not supply a solution.*

### Solution to Exercise 18.7

*The source does not supply a solution.*

### Solution to Exercise 18.8

*The source does not supply a solution.*

### Solution to Exercise 18.9

*The source does not supply a solution.*

# 19 Subsets of the complex plane

The modulus and real and imaginary parts describe curves and regions in the complex plane.

Visualize the following subsets of $\mathbb{C}$. Decide which sets are open, closed, or neither.

## 19.1 Exercises

**Exercise 19.1.** $\{z \mid |z-1| = 1\}$

**Exercise 19.2.** $\{z \mid |\mathrm{i}z-1| < 1\}$

**Exercise 19.3.** $\{z \mid |z-\mathrm{i}| + |z+\mathrm{i}| < 4\}$

**Exercise 19.4.** $\{z^2 \mid |z-\mathrm{i}| = 1\}$

**Exercise 19.5.** $\{z \mid \Re(e^{\mathrm{i}\pi/2}z) > 0\}$

**Exercise 19.6.** $\{z \mid \Im(z^2) > 0\}$

## 19.2 Solutions

### Solution to Exercise 19.1

Square the equation and introduce $x$ and $y$ to get the equation for a circle, i.e., boundary of a circle. Alternatively just realize that the modulus $|w|$ is the distance from the origin.

### Solution to Exercise 19.2

Interior of a circle.

### Solution to Exercise 19.3

Ellipse with foci at $\pm\mathrm{i}$.

### Solution to Exercise 19.4

Parameterize the original circle by $z(t) = \mathrm{i} + e^{\mathrm{i}t}$, $0 \le t < 2\pi$, and plot or analyze the image curve $z(t)^2$.

### Solution to Exercise 19.5

This is a half-plane.

### Solution to Exercise 19.6

*The source does not supply a solution.*

# 20 The complex exponential

The functional equation for the exponential and the Cauchy–Riemann equations determine the complex exponential.

In this section, we give some relatively easy exercises to illustrate the important concepts of complex analytic functions.

In this exercise, we derive the complex exponential function. The starting point is the real exponential function, which is the unique function $f(x) = \exp(x)$ that satisfies

$$f(x_1 + x_2) = f(x_1)f(x_2), \quad f(0) = 1.$$

We desire to generalize $\exp(x)$ to the complex plane, so that $\exp(x + 0\mathrm{i}) = \exp(x) \in \mathbb{R}$.

To avoid confusion, we call this unknown function $f : \mathbb{C} \to \mathbb{C}$.

## 20.1 Exercises

**Exercise 20.1.** Show that $f(x+\mathrm{i}y) = \exp(x)f(\mathrm{i}y)$. (Thus, we need to determine $f(\mathrm{i}y)$.)

**Exercise 20.2.** We set $f(\mathrm{i}y) = A(y) + \mathrm{i}B(y)$. Show that

$$A(y) = B'(y), \quad B(y) = -A'(y),$$

and hence that $A''(y) = -A(y)$. Hint: Cauchy–Riemann.

**Exercise 20.3.** The general solution to the ODE $A'' = -A$ is

$$A(y) = \alpha \cos(y) + \beta \sin(y),$$

where $\alpha$ and $\beta$ are constants to be determined. Show that $\alpha = 1$ and $\beta = 0$ are the only constants compatible with $f(z)$ being a generalization of the real exponential function.

**Exercise 20.4.** Conclude that

$$f(x + \mathrm{i}y) = e^x(\cos y + \mathrm{i}\sin y)$$

**Exercise 20.5.** Show that $(\exp(z))' = \exp(z)$, using Cauchy–Riemann, and that $\exp(z)$ is everywhere analytic.

**Exercise 20.6.** Show that the equation

$$\exp(z) = w$$

has infinitely many solutions for any $w \neq 0$.

**Exercise 20.7.** Show that $\exp(z) \neq 0$.

## 20.2 Solutions

### 20.2.1 Tutor note: The complex exponential function

**Tutor note:** I think that this exercise is very nice. It is not too hard if the student has some experience with the equation $y'' = -y$.

#### Solution to Exercise 20.1

*The source does not supply a solution.*

#### Solution to Exercise 20.2

*The source does not supply a solution.*

#### Solution to Exercise 20.3

*The source does not supply a solution.*

### Solution to Exercise 20.4

*The source does not supply a solution.*

### Solution to Exercise 20.5

*The source does not supply a solution.*

### Solution to Exercise 20.6

*The source does not supply a solution.*

### Solution to Exercise 20.7

*The source does not supply a solution.*

# 21 Complex power series

Geometric and exponential series give power-series representations of elementary complex functions.

## 21.1 Exercises

**Exercise 21.1.** Consider the geometric series

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \ldots \longrightarrow \frac{1}{1-z}.$$

Show that the $N$th partial sum is

$$\sum_{n=0}^{N} z^n = \frac{1 - z^{N+1}}{1-z}.$$

Polynomial long division may be useful.

**Exercise 21.2.** Find an upper bound for $|z|$ such that the partial sum converges to $1/(1-z)$.

**Exercise 21.3.** The Taylor series for the exponential function is

$$\sum_{n=0}^{\infty} \frac{1}{n!} z^n.$$

Using the definition of the complex exponential, find the Taylor series for $\cos(x)$ and $\sin(x)$, for $x \in \mathbb{R}$.

**Exercise 21.4.** Make plots of the partial sums of the sine and cosine Taylor series to verify your result.

**Exercise 21.5.** Find the Taylor series for the function $f(z) = (e^z - 1)/z$

**Exercise 21.6.** Find the Taylor series for the function $f(z) = \sin(z)/z$ for $z \neq 0$ and $f(0) = 1$.

## 21.2 Solutions

### Solution to Exercise 21.1

*The source does not supply a solution.*

### Solution to Exercise 21.2

*The source does not supply a solution.*

### Solution to Exercise 21.3

*The source does not supply a solution.*

### Solution to Exercise 21.4

*The source does not supply a solution.*

### Solution to Exercise 21.5

*The source does not supply a solution.*

### Solution to Exercise 21.6

*The source does not supply a solution.*

# 22 Singularities and contour integrals

Laurent series classify isolated singularities and determine contour integrals around poles.

## 22.1 Exercises

### 22.1.1 Singularities

**Exercise 22.1.** Consider the function $f(z) = \frac{1}{z(z-1)}$. How many poles does $f$ have, and where are they? What are the order of the poles? Can you write down the Laurent series of $f$ around $z = 0$?

**Exercise 22.2.** Does the function $f(z) = 1/\sin(z)$ have a pole? Where? What order?

**Exercise 22.3.** Find the singularities of $f(z) = e^{-1/(z-1)^2}$.

### 22.1.2 Line integrals

Let

$$f(z) = \frac{1}{z^3 - (2+\mathrm{i})z^2 + (1+2\mathrm{i})z - \mathrm{i}}.$$

**Exercise 22.4.** Let $\Gamma_\epsilon(w)$ be the simple closed curve defined by the counter-clockwise border of the $\epsilon$-ball $B_\epsilon(w)$. Write down a parameterization (a path) for this curve.

**Exercise 22.5.** Identify the poles $P = \{w_1, w_2, \cdots\}$ of $f(z)$ and their order. Hint: A zero of the denominator is $z = \mathrm{i}$.

**Exercise 22.6.** What is the value of

$$\oint_{\Gamma_\epsilon(0)} f(z)\,\mathrm{d}z$$

for $\epsilon = 1/2$?

**Exercise 22.7.** Find the value of the line integrals

$$\oint_{\Gamma_\epsilon(w_i)} f(z)\,\mathrm{d}z$$

for $\epsilon = 1/2$. Hint: Do not attempt the integral directly; compute the relevant Laurent coefficient. It can be useful to use the Taylor expansion

$$\frac{1}{a-z} = \frac{1}{a}\frac{1}{1-z/a} = \frac{1}{a}\sum_{n=0}^{\infty}(z/a)^n.$$

## 22.2 Solutions

### Solution to Exercise 22.1

The poles are simple and occur at $z = 0$ and $z = 1$. For $0 < |z| < 1$,

$$f(z) = \frac{1}{z(z-1)} = -\frac{1}{z(1-z)} = -z^{-1} - 1 - z - z^2 - \cdots.$$

### Solution to Exercise 22.2

The zeros of $\sin z$ occur at $z = k\pi$, $k \in \mathbb{Z}$, and each zero is simple because $\cos(k\pi) = (-1)^k \neq 0$. Thus $1/\sin z$ has a simple pole at every $k\pi$. Near the origin, $1/\sin z \sim 1/z$.

### Solution to Exercise 22.3

The exponent has a pole at $z = 1$ (of order 2). This means that the exponential function gives an essential singularity at $z = 1$.

### Solution to Exercise 22.4

$\gamma(t) = w + \epsilon e^{it}, 0 \leq t < 2\pi.$

### Solution to Exercise 22.5

The denominator can be factorized as $(z-1)^2(z-\mathrm{i})$, and therefore we have a pole of order 2 at $w_1 = 1$ and a pole of order 1 at $w_2 = \mathrm{i}$.

### Solution to Exercise 22.6

The function has no poles inside the $\epsilon$-ball, so the integral is zero by Cauchy's integral theorem.

### Solution to Exercise 22.7

Since

$$f(z) = \frac{1}{(z-1)^2(z-\mathrm{i})},$$

the residue at the second-order pole $z = 1$ is

$$\operatorname{Res}(f, 1) = \frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{z-\mathrm{i}}\bigg|_{z=1} = -\frac{1}{(1-\mathrm{i})^2} = -\frac{\mathrm{i}}{2}.$$

The residue at the simple pole $z = \mathrm{i}$ is

$$\operatorname{Res}(f, \mathrm{i}) = \frac{1}{(\mathrm{i}-1)^2} = \frac{\mathrm{i}}{2}.$$

Therefore

$$\oint_{\Gamma_{1/2}(1)} f(z)\,\mathrm{d}z = \pi, \qquad \oint_{\Gamma_{1/2}(\mathrm{i})} f(z)\,\mathrm{d}z = -\pi.$$

# Part IX

# Numerical methods

# 23 Dual numbers and automatic differentiation

Dual numbers adjoin a nonzero element $\varepsilon$ with $\varepsilon^2 = 0$, which encodes first derivatives algebraically.

We now consider the number system called *the dual numbers*, often written $\mathbb{R}(\varepsilon)$. The dual numbers form an algebraic extension of $\mathbb{R}$ with a nonzero element $\varepsilon$ such that $\varepsilon^2 = 0$:

$$\mathbb{R}(\varepsilon) = \{x + \varepsilon y \mid x, y \in \mathbb{R}\}.$$

Their construction resembles $\mathbb{C} = \mathbb{R}(\mathrm{i})$, but $\mathrm{i}^2 = -1$ while $\varepsilon^2 = 0$. Dual numbers support automatic differentiation.

## 23.1 Exercises

**Exercise 23.1.**

1. Compute the multiplication law in terms of real and "dual" parts, i.e., $z_1 z_2 = z = x + \varepsilon y$, and find $x$ and $y$. Compute the addition law for $z_1 + z_2$.
2. Show that $\mathbb{R}(\varepsilon)$ is not a field.
3. Compute the law for all positive powers of $z = x + \varepsilon y$, and for negative powers when $x \neq 0$.
4. Consider $z = x + \varepsilon$ and evaluate its positive powers, and its negative powers when $x \neq 0$. How is each dual part related to the corresponding real part?
5. Let $p(x)$ be a real polynomial and extend it to dual arguments. Show that

$$p(x + \varepsilon) = p(x) + \varepsilon p'(x).$$

6. Let $q(x)$ be a polynomial in negative powers of $x$. Show, for $x \neq 0$, that $q(x+\varepsilon) = q(x) + \varepsilon q'(x)$.
7. Let $p(x)$ and $q(x)$ be polynomials, and consider the *rational function* $f(x) = p(x)/q(x)$. Where $q(x) \neq 0$, extend the rational function to $\mathbb{R}(\varepsilon)$ and prove that $f(x+\varepsilon) = f(x) + \varepsilon f'(x)$.

## 23.2 Solutions

### 23.2.1 Tutor note: Dual numbers

This exercise is perhaps for the specially interested.

#### Solution to Exercise 23.1

**For part 2.**

Since $\varepsilon^2 = 0$, $\varepsilon$ cannot have an inverse. If $a\varepsilon = 1$, multiplication by $\varepsilon$ would give $a\varepsilon^2 = \varepsilon$, hence $0 = \varepsilon$, a contradiction.

**For part 3.**

Use the binomial formula and $\varepsilon^2 = 0$.

**For part 4.**

$(x+\varepsilon)^n = x^n + \varepsilon n x^{n-1}$. The dual part is the derivative of the real part.

$$(x+\varepsilon)^{-1} = \frac{x-\varepsilon}{(x-\varepsilon)(x+\varepsilon)} = \frac{x-\varepsilon}{x^2} = x^{-1} - \varepsilon x^{-2}.$$

Computing further,

$$(x+\varepsilon)^{-n} = x^{-n} - \varepsilon n x^{-n-1}.$$

Again, the dual part is the derivative of the real part.

# 24 Newton's method

Newton's method replaces a nonlinear system by its first-order Taylor approximation at each iteration.

In the next exercise, we derive the Newton–Raphson method for root finding.

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$ be a function of class $C^2$. We seek $\mathbf{x} \in \Omega$ such that $f(\mathbf{x}) = 0$.

Assume we have an initial guess $\mathbf{x}_0 \in \Omega$.

## 24.1 Exercises

**Exercise 24.1.** Explain why the following polynomial is a good approximation to $f(\mathbf{x})$ near $\mathbf{x}_0$:

$$p(\mathbf{x}) = f(\mathbf{x}_0) + J(\mathbf{x} - \mathbf{x}_0), \quad J = Df(\mathbf{x}_0).$$

**Exercise 24.2.** Newton's method now finds a new guess $\mathbf{x}_1$ by finding the root of the Taylor polynomial, i.e., $p(\mathbf{x}_1) = 0$. Write down a formula for $\mathbf{x}_1$, assuming that the matrix $J$ has an inverse.

**Exercise 24.3.** Suppose the true solution is $\mathbf{x}_* \in \Omega$, and assume that the error in $\mathbf{x}_0$ is sufficiently small. Find an estimate for the error of $\mathbf{x}_1$. Assuming that the error can be converted to significant digits $n$, what is the number of significant digits in $\mathbf{x}_1$? Hint: Look at the remainder term.

**Exercise 24.4.** Suppose successive iterations are performed. How many digits do you have after $k$ iterations?

**Exercise 24.5.** Make a Python implementation of Newton's method and test it on the following function:

$$f(x, y) = (e^x y^3 - 1, y^2 - \sin(x) - 1).$$

It can be an idea to try and visualize the function. Two roots are:

$$\{(4.38277027, 0.23202192),\ (0, 1)\}$$

Try different initial guesses $\mathbf{x}_0$, close to the roots and further away. How many iterations do you need to achieve machine precision for your guesses? Can you find more roots?

## 24.2 Solutions

### 24.2.1 Tutor note: Newton–Raphson method

**Tutor note:** Newton–Raphson is an important nonlinear root-finding algorithm, although quantum-chemistry calculations may avoid a full Newton step because solving with the Jacobian is expensive. This exercise also develops Taylor approximations of vector-valued functions.

#### Solution to Exercise 24.1

For a $C^2$ function, Taylor's theorem gives

$$f(\mathbf{x}) = f(\mathbf{x}_0) + Df(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + \mathcal{O}(\|\mathbf{x} - \mathbf{x}_0\|^2).$$

#### Solution to Exercise 24.2

$$\mathbf{x}_1 = \mathbf{x}_0 - J^{-1} f(\mathbf{x}_0).$$

#### Solution to Exercise 24.3

*The source does not supply a solution.*

## Solution to Exercise 24.4

*The source does not supply a solution.*

## Solution to Exercise 24.5

*The source does not supply a solution.*

# 25 Complex-step differentiation

Complex-step differentiation estimates derivatives without the subtractive cancellation of a centered finite difference.

Let $f(x)$ be a real-valued function, assumed to be analytic near some $x$, i.e., it agrees with its Taylor series in the vicinity of $x$.

## 25.1 Exercises

**Exercise 25.1.**

1. Explain why there exists a complex analytic function near $x + 0\mathrm{i}$ that agrees with $f$ on the real axis.

A classical method for computing a numerical derivative of a $C^1$ function is the finite difference approximation:

$$f'(x) \approx \delta_h f(x) := \frac{f(x+h) - f(x-h)}{2h}.$$

The error is $\mathcal{O}(h^2)$. In the rest of the exercise, let $f$ be given by

$$f(x) = \exp(x),$$

and we wish to differentiate around $x = 1$.

1. Write a small program that uses standard double precision floating point arithmetic to compute the finite difference derivative for step lengths from the interval $h \in [10^{-9}, 10^{-1}]$.
2. Make a plot of the absolute value of the error in the approximation as function of $h$. Use log scales.

The *complex step method* utilizes complex analyticity to avoid the cancellation errors seen in the finite difference scheme. It relies on the function in question being implemented/implementable using complex arithmetic.

1. Show that

$$f(x + \mathrm{i}h) = f(x) + \mathrm{i}hf'(x) - \frac{1}{2}h^2 f''(x) - \frac{\mathrm{i}h^3}{6} f'''(x) + \mathcal{O}(h^4).$$

2. Solve for $f'(x)$, and show that

$$f'(x) = \Im \frac{f(x + \mathrm{i}h)}{h} + \mathcal{O}(h^2).$$

   This is the complex step method.

3. In the plot from above, add a plot of the error of the complex step method. Compare.

## 25.2 Solutions

### 25.2.1 Tutor note: Complex step method

**Tutor note:** I think the tutors might be surprised by this exercise! I learned about the method only last year. This exercise is very nice, in my opinion. However, it may be a little advanced for some students, because it requires basic understanding of cancellation errors in floating point arithmetic.

#### Solution to Exercise 25.1

**Source solution 1.**

The real power series has a positive radius of convergence. Using the same coefficients for a complex argument gives a complex power series with the same radius of convergence, and hence a unique analytic extension near $x$.

**Source solution 2.**

The student should observe that the accuracy deteriorates when $h$ becomes too small, around $10^{-7}$.

**Source solution 3.**

Just compute the Taylor series.

**Source solution 4.**

The main observation is that the zeroth and second order terms are purely real, while the first and third are purely imaginary. The given approximation then eliminates the real parts, which explains the second-order error.

**Source solution 5.**

The student should observe that the accuracy does *not* deteriorate as it does for the centered difference. The complex-step formula avoids subtraction of nearly equal function values.

# Part X

# Fourier analysis

# Part XI

# Ordinary differential equations

# Part XII

# Partial differential equations

# Part XIII

# Optimization